Active Portfolio Management (Grinold & Kahn, 2nd ed., 1999) — Quant Study Companion

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1. Book map and fundamental thesis

Thesis in one sentence: active management is forecasting residual returns, and the value a manager can add is governed by one number, the information ratio (IR), which decomposes as IR ≈ IC × √BR. Every other part of the book (risk models, alpha refinement, portfolio construction, trading, performance analysis) exists to measure, preserve, or verify that number.

The authors' own compression is the "Seven Quantitative Insights" talk listed in the Preface (p. xii), paraphrased:

1. Active management is forecasting; consensus views lead you back to the benchmark.
2. The information ratio is the key to value added.
3. The fundamental law: IR = IC × √BR.
4. Alphas must control for volatility, skill and expectations: alpha = volatility × IC × score.
5. Data mining is easy, so you need guidelines to avoid it.
6. Implementation should subtract as little value as possible.
7. Distinguishing skill from luck is difficult.

The four closing themes (Ch. 22, pp. 578–580) restate this: active management is a process; it is forecasting with superior information; forecast as often as possible (breadth); and mathematics cannot overcome ignorance.

Chapter map

Page numbers are the printed book pages (the PDF runs about 8 pages ahead).

Part	Ch	Title	Pages	Core question	Signature result
—	1	Introduction	1–8	Why a science of active management?	Benchmark-relative view; roadmap of the process

Part	Ch	Title	Pages	Core question	Signature result
I Foundations	2	Consensus Expected Returns: the CAPM	11–40	What does the market already expect?	Return = systematic + residual; CAPM says E[residual] = 0; characteristic portfolios (appendix)
I	3	Risk	41–86	How do we measure and forecast risk?	Risk = std. dev.; risks don't add; structural factor models; marginal contributions
I	4	Exceptional Return, Benchmarks, Value Added	87–108	What is the manager's objective?	Expected return = time premium + risk premium + exceptional benchmark return + alpha; VA = $\alpha - \lambda\omega^2$
I	5	Residual Risk and Return: the Information Ratio	109–146	What opportunities does a manager have?	Residual frontier; $\omega^* = IR / 2\lambda$; $VA^* = IR^2 / 4\lambda$; empirical IR distributions
I	6	The Fundamental Law of Active Management	147–172	Where does IR come from?	$IR \approx IC \times \sqrt{BR}$; additivity across sources
II Expected Returns and Valuation	7	Expected Returns and the APT	173–198	Can factors forecast returns?	APT is a model of the manager's, not the consensus, expected returns
II	8	Valuation in Theory	199–224	What is a stock worth?	Value = risk-adjusted expected cash flows at the risk-free rate; misvaluation adds a correction term to expected return
II	9	Valuation in Practice	225–258	How do practitioners value equities?	DDM (constant growth, three-stage), implied growth, comparative valuation, returns-based analysis
III Information Processing	10	Forecasting Basics	261–294	How do raw signals become alphas?	Basic forecasting formula; alpha = volatility × IC × score
III	11	Advanced Forecasting	295–314	Multi-asset, cross-sectional scores, factor forecasts	Cross-sectional scores often need no volatility scaling; IC uncertainty shrinks alpha
III	12	Information Analysis	315–346	Does a signal contain information?	Two steps: signal → portfolio → performance; event studies; data mining is easy

Part	Ch	Title	Pages	Core question	Signature result
III	13	The Information Horizon	347–376	How fast does information decay?	Horizon = half-life of IR; lagged signals and past returns can help
IV Implementation	14	Portfolio Construction	377–418	How do alphas become holdings?	Alpha analysis (scale, trim, neutralize); QP vs screens/stratification/LP; dispersion
IV	15	Long/Short Investing	419–444	What does the long-only constraint cost?	Benefits grow with universe size, low volatility, high active risk; long-only induces a small-cap bias
IV	16	Transactions Costs, Turnover, Trading	445–476	How much value does trading destroy?	Implementation shortfall; square-root impact; half the turnover keeps ⩾ ¾ of value added
IV	17	Performance Analysis	477–516	Skill or luck?	Returns regression and t-stat; style analysis; holdings-based attribution
IV	18	Asset Allocation	517–540	Active management with 3–20 assets	Three-step process; international consensus returns; currencies
IV	19	Benchmark Timing	541–558	Can you time the market?	One-dimensional bet: breadth only through frequency; futures beat stocks
IV	20	The Historical Record for Active Management	559–572	Does the evidence support the framework?	Average fund ≈ index net of fees; weak persistence; a population model
—	21	Open Questions	573–576	What the theory doesn't cover	Dynamics, t-cost models, liabilities, nonlinearity (size), taxes, behavioral finance
—	22	Summary	577–580	—	Four themes (above)
—	A–C	Notation, Glossary, Return and Statistics Basics	581–590	—	Reference material

Every chapter from 2 to 19 ends with a technical appendix (formal propositions and proofs) and applications exercises, which is where most of the mathematics lives.

2. The authors' assumptions and how the chapters depend on each other

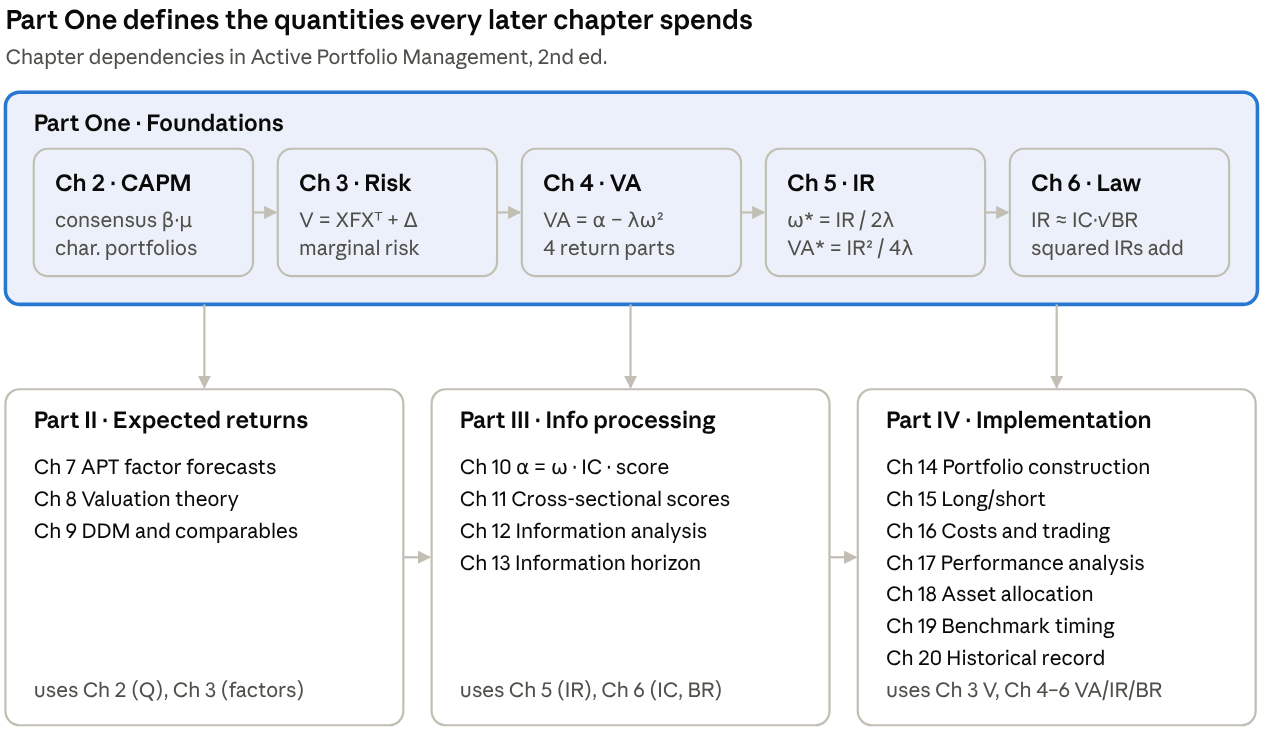
The framework rests on fourteen assumptions. Most are stated openly; the most important one, that the manager knows the true value of his or her own information, is the least testable and is flagged by the authors themselves (p. 162).

#	Assumption	Where stated	What depends on it	When it bends
1	Markets are "somewhat efficient": the burden of proof sits with the manager (humility principle)	pp. 2, 23	The whole posture of the book; consensus as the default	Never fully, but it is a prior, not a theorem
2	Consensus expected returns are those that make the benchmark mean/variance efficient	p. 18; (2A.36)	Definition of alpha as deviation from consensus	Benchmark is not on the true efficient frontier (Roll critique)
3	Performance is judged relative to a benchmark; risk is symmetric	pp. 4–5, 42	Active/residual risk focus; tracking error as the risk measure	Absolute-return mandates, drawdown-constrained capital
4	Risk = standard deviation; cost of risk ∝ variance (normal returns, quadratic utility, or short horizon)	p. 47; p. 96 fn. 3	Mean/variance objective, value added = $\alpha - \lambda\omega^2$	Options, dynamic strategies, crash-prone factor bets (p. 47 fn. 4)
5	Returns are serially uncorrelated, so variance scales with time	p. 49; Prop. 5, p. 143	Annualizing risk; IR ∝ $\sqrt{T}$	Momentum, reversal, illiquid or smoothed returns
6	Factor exposures are known at the start of each period; specific returns are uncorrelated with factors and with each other	p. 57; (3A) A1–A2	Structural risk models, $V = XFX^T + \Delta$	Missing factors, crowded trades, thin industries (p. 80 fn. 14)

#	Assumption	Where stated	What depends on it	When it bends
7	Risk forecasting can be separated from return forecasting	p. 4 and fn. 3	Using a vendor risk model with your own alphas	Alpha factors that are also risk factors (alpha/risk misalignment)
8	The client bears benchmark risk; the manager bears active risk	pp. 100–101	Separate aversions λ_T, λ_{BT}, λ_R; the value-added objective	Sponsors who care about total risk or drawdowns
9	The IR does not depend on aggressiveness	p. 116	Residual frontier as a straight line; $\omega^* = IR / 2\lambda$	Long-only and position constraints, capacity and costs
10	Alphas are benchmark-neutral; benchmark timing is set aside	pp. 101–102, 112	Focus on residual return; $\beta = 1$ portfolios	Managers who do time markets (Ch. 19)
11	The manager accurately gauges his or her own skill and uses it optimally ("hypercompetence")	pp. 159, 162	Fundamental law; all optimal-policy results	Almost always: ICs are estimated with large error and decay
12	Signals are independent, equally skilled, and weak (IC below about 0.1)	pp. 148, 157–159	$IR \approx IC\sqrt{BR}$ and its additivity	Repackaged signals, factor crowding, cross-sectional correlation
13	Single period, frictionless; costs and constraints come later	pp. 27, 109	Clean closed forms in Part One	Turnover-heavy strategies (Ch. 14–16)
14	Mathematics cannot create information; it only uses it efficiently	p. 579	The limits of the whole machinery	—

Read the table as a stack: rows 1–7 are classical portfolio theory the authors borrow, rows 8–10 are the institutional adaptation that defines "active" management, and rows 11–13 are the simplifications that make the fundamental law work. The diagram below shows how

the chapters build on each other.



chapter dependencies · Part One feeds Parts II–IV

Part Two produces raw return forecasts, Part Three refines them into alphas, and Part Four turns alphas into traded and audited portfolios; each part spends the quantities Part One defines.

3. Reading and research plan

The book is worked through in six installments. This first one covers the whole-book map plus Part One (Ch. 1–6), because every later chapter reuses Part One's objects: benchmark, alpha, residual risk, IR, IC and breadth.

Method. The PDF's prose is machine-readable, but its equations are embedded as images (about 1,000 of them). Prose is read from extracted text; every equation cited here is checked against the rendered page image. One PDF page (the Figure 17.3 page, PDF p. 490) carries a wrong "Page 4" label; it was re-filed under Ch. 17 so it does not overwrite the real p. 4.

1. **Installment 1 (this one):** map, thesis, assumptions, dependency graph; Ch. 1–6 in full, including technical appendices; tested Python toolkit for Part One; first mental models, strategy entries and experiments.

2. **Installment 2, Part Two (Ch. 7–9):** APT, valuation theory, dividend discount and comparative valuation models.
3. **Installment 3, Part Three (Ch. 10–13):** forecasting formula, cross-sectional scores, IC shrinkage, information analysis, event studies, information horizon. This is the core of a signal-research pipeline.
4. **Installment 4, Part Four (Ch. 14–16):** alpha analysis and portfolio construction, long/short, transaction costs and trading.
5. **Installment 5 (Ch. 17–22 and appendices):** performance analysis, asset allocation, benchmark timing, the historical record, open questions.
6. **Installment 6, synthesis:** consolidated math reference, research-system design, the "challenge the author" review against later academic work, modern-methods integration, comprehensive exam, and the final ranked synthesis.

Each chapter installment follows the same template: thesis, concepts, derivations with numerical examples, trading implications, code, critique, takeaways, then five conceptual questions, three math exercises with solutions, two coding exercises and one research experiment.

Coverage tracker

Chapter	Pages	Installment	Coverage
1 Introduction	1–8	1	Full text + equations checked
2 CAPM	11–40	1	Full text + equations checked
3 Risk	41–86	1	Full text + equations checked
4 Exceptional Return, Benchmarks, Value Added	87–108	1	Full text + equations checked
5 Information Ratio	109–146	1	Full text + equations checked
6 Fundamental Law	147–172	1	Full text + equations checked
7 APT	173–198	2	Headings, opening, summary
8 Valuation in Theory	199–224	2	Headings, opening, summary

Chapter	Pages	Installment	Coverage
9 Valuation in Practice	225–258	2	Headings, opening, summary
10 Forecasting Basics	261–294	3	Headings, opening, summary
11 Advanced Forecasting	295–314	3	Headings, opening, summary
12 Information Analysis	315–346	3	Headings, opening, summary
13 Information Horizon	347–376	3	Headings, opening, summary
14 Portfolio Construction	377–418	4	Headings, opening, summary
15 Long/Short Investing	419–444	4	Headings, opening, summary
16 Transactions Costs, Turnover, Trading	445–476	4	Headings, opening, summary
17 Performance Analysis	477–516	5	Headings, opening, summary
18 Asset Allocation	517–540	5	Headings, opening, summary
19 Benchmark Timing	541–558	5	Headings, opening, summary
20 Historical Record	559–572	5	Headings, opening, summary
21 Open Questions	573–576	5	Full text read
22 Summary	577–580	5	Full text read
Appendices A–C, Index	581–597	5	Not yet examined

4. Mental model library (Part One)

Fourteen models come out of Part One; later installments will add models from valuation, forecasting and implementation. Each carries a status tag: **[Math]** is an identity that holds given its stated inputs, **[Stats]** is an established statistical result, **[Premise]** is an institutional or economic assumption, and **[Hypothesis]** needs empirical verification on your own data.

M1. Consensus in, benchmark out [Math + Premise]

- **Core:** any portfolio you call optimal implies a unique set of expected returns, $f = f_B \times V h_B / \sigma_B^2$. If your forecasts equal those, the optimizer returns the benchmark (p. 18, (2A.36)).
- **Intuition:** an optimizer is a translator between beliefs and holdings. Feed it the market's beliefs and it hands back the market.
- **Market mechanism:** cap-weighted holdings aggregate everyone's positions, so they are the natural "average belief".
- **Trading use:** reverse-optimize the benchmark to get a neutral prior; express only the deviations as alpha. This is the backbone of Black–Litterman (outside the book).
- **Fails when:** the benchmark is not efficient (most real indices), or the covariance matrix is wrong, so implied returns inherit risk-model error.
- **Connects to:** M2, M8, Ch. 7 (APT as the manager's, not the consensus, model).

M2. Every return splits into a beta part and a residual part [Math]

- **Core:** $r_P = \beta_P r_B + \theta_P$, with θ_P uncorrelated with r_B; no theory is needed for this (p. 16). The CAPM adds only the claim $E[\theta] = 0$.
- **Intuition:** project each return onto the benchmark and keep what is left.
- **Trading use:** hold $\beta = 1$ and research only residual returns, whose consensus forecast is zero (p. 23).
- **Fails when:** betas are unstable; historical betas regress toward 1 (p. 14 fn. 4), so residuals estimated with stale betas leak market exposure.
- **Connects to:** M3, M7, Ch. 3 factor models (a multi-factor generalization).

M3. Residual returns are a zero-sum game before costs [Math]

- **Core:** cap-weighted residual returns sum to exactly zero (p. 16, (3.14)). For every overweight there is someone underweight (the investor A/B/C story, p. 17).

- **Intuition:** you can only beat the benchmark by taking money from other active investors.
- **Market mechanism:** after fees and trading costs, the average active dollar must lag. Ch. 16 cites Treynor's version of this argument (p. 445).
- **Trading use:** always ask who is on the other side and why they are wrong. Edges come from better information, better processing, or different constraints and time horizons of the counterparty.
- **Connects to:** the symmetric IR distribution centred on zero in Table 5.1 (p. 114); Ch. 20.

M4. Risks don't add; diversification has a floor [Math]

- **Core:** with N stocks of risk σ and pairwise correlation ρ, an equal-weight portfolio has risk $\sigma\sqrt{((1 + \rho(N-1))/N)}$, which tends to $\sigma\sqrt{\rho}$ (3.4)–(3.5).
- **Example:** 35% stocks with ρ = 0.5 give 27.1% at N = 5, 25.4% at N = 20, and a floor of 24.7%.
- **Trading use:** residual bets diversify (ρ near 0); common-factor bets do not. Breadth only helps for the part of risk that is uncorrelated.
- **Connects to:** M9 (breadth is the diversification of residual bets), Ch. 3.

M5. Structure beats sample: errors concentrate where the optimizer looks [Stats]

- **Core:** an N × N sample covariance from T periods has rank min(N, T−1); with T ⩽ N some active positions look riskless (p. 54, fn. 7). Structural models reduce the problem to K factors plus N specific variances.
- **Evidence here:** in experiment E4 (synthetic, Section 12), optimized portfolios built on a 120-month sample covariance realized 5.6× their predicted risk; the estimated factor model was off by 1.05×. Random portfolios were fine under both (1.02–1.03×).
- **Fails when:** the factor set misses a real common driver, or exposures drift between estimation dates.
- **Connects to:** M6, Ch. 14 (optimizers amplify alpha and risk errors alike).

M6. Marginal contribution equals beta to the portfolio; optimality means alpha ∝ MCRR [Math]

- **Core:** MCTR = V h_P / σ_P (3A.20) and h_P^T MCTR = σ_P. At the optimum, α = IR × MCRR_Q (5A.15).

- **Intuition:** at the optimum, every position's marginal return per unit of marginal risk is the same, and that common ratio is the IR.
- **Trading use:** regress your alphas on the portfolio's marginal contributions to residual risk. The slope is the IR your holdings imply; a low R^2 means the book does not express your views (p. 136). The toolkit's `alpha_mcrr_check` does this.
- **Connects to:** M8, Ch. 3 risk attribution, Ch. 14.

M7. Business risk versus investment risk [Premise]

- **Core:** the sponsor owns benchmark risk, the manager owns active risk, so they get separate prices: $VA = (\beta_{PA}\,\Delta f_B - \lambda_{BT}\,\beta_{PA}^2\,\sigma_B^2) + (\alpha_P - \lambda_R\,\omega_P^2)$ (4.15).
- **Intuition:** going from 20% to 21% total risk looks trivial, but it carries 6.4% residual risk, which decides who ends up bottom of the league table (p. 100).
- **Trading use:** calibrate λ_R from a target tracking error: $\lambda_R = IR / (2\omega^*)$ (5.11).
- **Fails when:** the mandate is absolute return, or the capital is drawdown-sensitive (most hedge funds and prop desks).
- **Connects to:** M8, Ch. 19 (benchmark timing).

M8. The IR is a budget constraint; value added grows with its square [Math]

- **Core:** along the residual frontier $\alpha = IR \times \omega$ (5.7). With $VA = \alpha - \lambda\omega^2$, the optimum is $\omega^* = IR / 2\lambda$ and $VA^* = IR^2 / 4\lambda$ (5.10)–(5.12).
- **Intuition:** risk is the fuel and IR is the engine's efficiency. Doubling IR quadruples what you can deliver at the same risk aversion.
- **Trading use:** rank strategies and managers by IR, not by return; every investor agrees on that ranking whatever their risk aversion (p. 125).
- **Fails when:** constraints, capacity or costs make IR fall as risk rises, so the frontier bends (p. 116; experiment E3).
- **Connects to:** M9, M13.

M9. The fundamental law: play often, play well [Math, approximation]

- **Core:** $IR \approx IC \times \sqrt{BR}$ (6.1); value added $\approx IC^2 \times BR / 4\lambda$ (6.3).
- **Intuition:** each independent bet is a weak coin flip; many independent coin flips diversify into a reliable edge, as in the casino example (p. 151).
- **Evidence here:** in experiment E1, realized IRs of optimal portfolios matched $IC\sqrt{BR}$ within one standard error across six IC × N settings.

- **Trading use:** a 50% increase in breadth is worth the same as a 22% increase in skill (p. 149). Use it to price research projects before running them.
- **Fails when:** bets are not independent (M10), skill is not uniform, or the manager mis-estimates IC (assumption 11).
- **Connects to:** M10, M11, M12, Ch. 18–19 (why timing struggles).

M10. Breadth must be independent [Math]

- **Core:** two signals of equal IC with correlation γ combine to IC(com) = IC√(2/(1+γ)) (6.9). Rebalancing monthly on an annual view is one bet made 12 times (p. 158).
- **Evidence here:** in E2 (six seeds), γ = 0.9 gave a realized IR of 1.83 against 2.53 if the two signals were counted as independent.
- **Trading use:** regress signals on industries and style factors; only the residuals are new bets (p. 158). Measure effective breadth, not the count of names.
- **Connects to:** M9, Ch. 10–12.

M11. Tiny edges are enough [Math]

- **Core:** for ±1 forecasts, IC = 2 × (hit rate) − 1 (6.6). An IC of 0.02 is a 51% hit rate, yet over 200 stocks each quarter it gives IR ≈ 0.57 (p. 154).
- **Trading use:** do not discard signals because their ICs look small; discard them because they are unstable, costly to trade, or redundant.
- **Fails when:** costs exceed the edge; footnote 4 on p. 153 shows 0.8% round-trip costs cutting an IR of 1.16 to 0.70.

M12. Alpha = volatility × IC × score [Math under joint normality]

- **Core:** a standardized signal becomes a residual-return forecast by multiplying by residual volatility and by the IC (p. 7; the derivation comes in Ch. 10). The synthetic forecasts in Ch. 6, fn. 8 have exactly this form.
- **Intuition:** the forecast shrinks toward the consensus (zero) in proportion to how little you know. With IC = 0, "garbage in" becomes zeros out.
- **Trading use:** never feed raw scores or analyst targets to an optimizer; refine them first, then scale so the implied IR is plausible (Prop. 6, p. 144).
- **Connects to:** M8 (alpha scaling), Ch. 10, 11, 14.

M13. Constraints are a tax on IR [Math; size is empirical]

- **Core:** forbidding active beta and active cash gives $IR_Y = IR\sqrt{(1 - Corr^2(\theta_Q, \theta_C))}$ (5A.40). Long-only constraints cost more and tilt toward small caps (Ch. 15 preview).
- **Evidence here:** in E3, long-only portfolios fell from IR 0.98 to 0.75 at a 5% risk target, ex-ante tracking error reached only 2.6%, and active weights correlated −0.30 with log market cap.
- **Trading use:** measure the IR cost of each constraint before accepting it.
- **Connects to:** M8, Ch. 14–15.

M14. Skill is hard to tell from luck [Stats]

- **Core:** the ex-post IR is about the alpha t-statistic divided by $\sqrt{(\text{years})}$ (p. 112). A true IR of 0.5 needs about 16 years to reach t = 2.
- **Trading use:** judge a new strategy by the plausibility of its mechanism and its out-of-sample IC stability, not by a short track record.
- **Connects to:** Ch. 12, 17, 20.

5. Chapter 1 — Introduction (pp. 1–8)

A. Central thesis

Active management can be run as an engineering process rather than an art: forecast residual returns, convert them into portfolios with explicit risk control, trade efficiently, and audit the results. The process cannot create an edge; it only stops a real edge from being wasted (p. 8: "If it is garbage in, then it's garbage out; we can only help to process the garbage more effectively").

B. Key concepts and how they relate

- **Three camps (p. 2):** people who think beating the market is impossible, easy, or difficult. The authors place themselves in the third and put the burden of proof on the strategy.
- **Dirty data, flawed models, nonstationarity (p. 2):** experiments cannot be repeated, and real trading moves prices. This is why the book prefers robust rules of thumb over precise optimization.
- **Separating risk forecasting from return forecasting (p. 4):** risk is forecastable and can be bought from a vendor; returns are where the manager competes.

- **Benchmark-relative management (pp. 4–5):** benchmarks let sponsors pigeonhole managers, instruct them, and manage the aggregate. Setting the benchmark to cash recovers ordinary total-return analysis.
- **Characteristics are portfolios (p. 5):** every asset attribute maps to a portfolio (made precise in Ch. 2′s appendix), so heterogeneous signals can be studied with one toolkit.
- **The chain of ideas (pp. 5–8):** residual return → information ratio → value added ∝ IR² → fundamental law (IC and breadth) → valuation and forecasting → alpha = volatility × IC × score → information analysis → portfolio construction and trading → performance analysis.

C. Mathematical foundations

Chapter 1 only previews three results that later chapters derive:

$$\mathrm{VA}^* = \frac{\mathrm{IR}^2}{4\lambda_R}, \qquad \mathrm{IR} \approx \mathrm{IC}\sqrt{\mathrm{BR}}, \qquad \alpha_n = \omega_n \cdot \mathrm{IC} \cdot z_n$$

A quick sanity check shows why the book stresses humility. A 2,000-bet-per-year strategy (500 stocks, quarterly) with IC = 0.05 implies IR ≈ 2.2, more than double the 90th-percentile IR of 1.0 in Table 5.1 (p. 114), where the top quartile sits near 0.5 before fees. So either real ICs are closer to 0.01–0.02, or effective breadth is far smaller than the count of names. Both are true in practice, as Ch. 6 explains.

D. Trading implications

1. Build a process, not a collection of stock ideas. Every idea must pass through the same refine → optimize → trade → attribute pipeline.
2. Prefer breadth. Skepticism about benchmark timing follows directly from the fundamental law (p. 6).
3. Never optimize raw signals. Refine them into alphas first (p. 7).
4. Measure information content before backtesting (p. 7; detailed in Ch. 12).

E. Practical implementation

The book′s process maps onto a small set of interfaces. This skeleton is the architecture the later chapters fill in.

```
from dataclasses import dataclass
import numpy as np

@dataclass
class Universe:            # point-in-time: assets, benchmark weights,
residual vols
    ids: list; h_B: np.ndarray; omega: np.ndarray

class Signal:              # Ch. 9-10: raw information -> standardized score
z (mean 0, std 1)
    def score(self, asof) -> np.ndarray: ...

class Refiner:             # Ch. 10-11: alpha = omega * IC * z, benchmark-
neutralized
    def alpha(self, z, universe, ic) -> np.ndarray:
        a = universe.omega * ic * z
        return a - (a @ universe.h_B) * np.ones_like(a)   # crude
neutralization (beta ~ 1)

class RiskModel:           # Ch. 3: V = X F X' + Delta
    def cov(self, asof) -> np.ndarray: ...

class Optimizer:           # Ch. 5, 14: max alpha'h - lambda h'Vh - costs,
subject to constraints
    def target(self, alpha, V, h_now, lam) -> np.ndarray: ...

class Trader:              # Ch. 16: schedule trades to minimize impact +
risk
    def execute(self, h_now, h_target): ...

class Attribution:         # Ch. 17: realized active return -> factors,
specific, timing, costs
    def explain(self, h, returns): ...
```

F. Critical evaluation

- The perspective is a 1990s institutional long-only manager with a BARRA-style vendor risk model; both authors worked at BARRA (Acknowledgments, p. xv). Hedge-fund concerns such as leverage, financing, drawdowns and capacity get little space.
- "Superior information exists" is assumed, not demonstrated. Chapter 20 offers only mixed evidence.

- The "three camps" framing is rhetoric. It does not tell you how to estimate whether your own edge is real; Ch. 12 and 17 do that.
- The process view is the book's lasting strength. Its separation of signal, risk, construction and costs is still how systematic equity shops are organized.

G. Key takeaways

1. The burden of proof is on the strategy, not on the market.
2. Manage relative to a benchmark; risk-free cash as the benchmark recovers classical theory.
3. The IR is the figure of merit; value added scales with its square.
4. IR comes from skill × √breadth, so many small independent edges beat a few big calls.
5. Process cannot create information, but it decides how much of it reaches the P&L.

Test your understanding

Conceptual questions

1. Why does the book separate the risk-forecasting problem from the return-forecasting problem, and what could go wrong if your alpha signals are themselves risk factors?
2. In what sense is benchmark-relative analysis "more general" than total-return analysis (p. 5)?
3. Why might a strategy with a mediocre IC still deserve capital?
4. What does "garbage in, zeros out" mean for alpha = volatility × IC × score?
5. Which of the book's process stages would you expect to destroy the most value in a modern US equity strategy, and why?

Math exercises (solutions below)

1. A manager has IR = 0.6 and residual risk aversion 0.10 (per %²). What residual risk and value added are optimal?
2. A signal covers 300 stocks, refreshed monthly, with IC = 0.015. What IR does the fundamental law predict, and what IC would quarterly refresh need for the same IR?
3. A stock has 30% residual volatility and a score of +1.5 from a signal with IC = 0.04. What is its alpha?

Solutions. (1) $\omega^* = 0.6 / 0.2 = 3.0\%$; $VA^* = 0.36 / 0.4 = 0.9\%$. (2) $IR = 0.015 \times \sqrt{3600} = 0.90$; quarterly gives BR = 1200, so $IC = 0.90 / \sqrt{1200} = 0.026$, assuming monthly refreshes really are independent. (3) $\alpha = 30\% \times 0.04 \times 1.5 = 1.8\%$ per year.

Coding exercises

1. Implement the `Refiner` above with proper beta-neutralization: $\alpha \to \alpha - \beta\,(h_B^T\,\alpha)$ (p. 128 recipe) and test that $h_B^T\,\alpha = 0$.
2. Write a function that takes a signal history and returns the realized monthly IC series and its mean, standard deviation and t-statistic.

Research experiment

Take any signal you already use. Compute its realized IC month by month, its IR in a simple optimal portfolio, and the IR the fundamental law predicts from the average IC and the number of names. The ratio of realized to predicted IR is a first estimate of how much breadth you are losing to correlation between bets.

Connections: this chapter is the table of contents for everything that follows; there are no earlier chapters to connect to.

6. Chapter 2 — Consensus Expected Returns: the CAPM (pp. 11–40)

A. Central thesis

The CAPM is used as a source of consensus expected returns, not as a belief. Expected returns proportional to beta are exactly the returns that make the market (or benchmark) optimal, so an active manager is, by definition, someone whose forecasts differ from beta × market premium. The machinery built to prove the CAPM (beta separation and characteristic portfolios) is what the rest of the book reuses (pp. 11–12).

B. Key concepts

- **Separation of return (pp. 13–16):** any portfolio's excess return splits into $\beta_P\, r_M$ plus an uncorrelated residual θ_P. "So far, no CAPM" (p. 16).
- **The CAPM claim (p. 16):** $E[\theta] = 0$ for every stock. Market risk is the "hot potato" someone must hold; residual risk is self-imposed and unrewarded.
- **Greater-fool logic (p. 17):** investor A's residual bets have an exact mirror image, investor B. If A expects to win, B must expect to lose.
- **CAPM versus market efficiency (pp. 17–18):** consistent, not identical. The CAPM argument rests on the zero-sum structure; efficient-markets theory rests on prices containing information.

- **Expected returns ↔ portfolios (p. 18):** CAPM returns fed into an optimizer give back the market; assuming the market is optimal and backing out returns gives the CAPM.
- **Ex ante vs ex post (pp. 19–20):** the security market line is a forecast. Realized returns scatter around an "ex post market line" that slopes down whenever the market lags cash.
- **Historical averages are poor forecasts (p. 12):** the standard error of a mean return is σ/√Y, whatever the sampling frequency.
- **Beta forecasting (pp. 14, 23–24):** historical betas regress toward 1; fundamental-attribute betas forecast better (Rosenberg 1985).
- **Technical appendix (pp. 26–39):** characteristic portfolios, portfolios C (minimum variance), B (benchmark), q and Q (maximum Sharpe), A (alpha), the efficient frontier, and the CAPM as the statement Q = M.

C. Mathematical foundations

1. Beta and the historical regression (2.1)–(2.2)

```
\beta_P = \frac{\operatorname{Cov}(r_P, r_M)}{\operatorname{Var}(r_M)},
\qquad r_P(t) = \alpha_P + \beta_P\, r_M(t) + \epsilon_P(t)
```

- *Variables:* r_P, r_M are excess returns over the risk-free rate; α_P, β_P are realized intercept and slope; ε_P is the period residual.
- *Derivation:* minimizing Σ(r_P − α − βr_M)² gives the normal equations; the slope is the sample covariance over the sample variance.
- *Intuition:* beta is how many units of market move the portfolio inherits.
- *Example (p. 21):* American Express vs the S&P 500, 60 months to Dec 1992: β = 1.21 (s.e. 0.24), residual −78 bp/month (s.e. 96 bp, t ≈ −0.81), residual std 7.05%/month (24.4% a year), R^2 = 0.31. Five years of data cannot tell this alpha from zero.
- *Assumptions:* stable beta over the window, no serial correlation, market proxy is the right benchmark.
- *Use:* hedge ratios, beta-neutral construction, and realized alpha for performance analysis (Ch. 17).

2. Return and risk separation (2.3)–(2.4)

```
r_P = \beta_P\, r_M + \theta_P, \qquad \sigma_P^2 = \beta_P^2\, \sigma_M^2 +
\omega_P^2
```

- *Derivation:* define $\theta_P = r_P - \beta_P r_M$. By construction $\mathrm{Cov}(\theta_P, r_M) = \mathrm{Cov}(r_P, r_M) - \beta_P \mathrm{Var}(r_M) = 0$, so the variances add.
- *Example (Problem 3, p. 24):* a stock with $\beta = 1.15$ and 35% volatility, market at 20%, has $\omega = \sqrt{(35^2 - 1.15^2 \cdot 20^2)} = 26.4\%$. Most single-stock risk is residual.
- *Use:* tells you how much of a position's risk your alpha must pay for.

3. The CAPM (2.5)

```
E\{r_P\} = \beta_P\, E\{r_M\} = \beta_P\, \mu_M
```

- *Assumptions (theorem, p. 38):* mean/variance investors, all assets included, common knowledge of expected returns, agreement on covariances, no costs or taxes.
- *Example (Problems 1–2):* Sears, $\beta = 1.05$. With $\mu_M = 7\%$, consensus excess return is 7.35%. If the market lags bills by 5%, the conditional expected excess return is −5.25%.
- *Use:* the neutral prior that every alpha is measured against.

4. Characteristic portfolios (2A.1)–(2A.4)

```
h_a = \frac{V^{-1} a}{a^{\top} V^{-1} a}, \qquad \sigma_a^2 = \frac{1}
{a^{\top} V^{-1} a}, \qquad a = \frac{V h_a}{\sigma_a^2}
```

- *Variables:* a is any attribute vector (beta, E/P, a signal); V is the asset covariance matrix.
- *Derivation:* minimize $\tfrac{1}{2} h^{\mathsf{T}} V h$ subject to $h^{\mathsf{T}} a = 1$. The Lagrangian condition $Vh = \theta a$ gives $h = \theta V^{-1} a$; the constraint fixes $\theta = 1/(a^{\mathsf{T}} V^{-1} a)$.
- *Intuition:* the third identity says every asset's beta with respect to h_a equals its attribute. The attribute and the portfolio are the same object seen from two sides.
- *Example:* two assets with 20% and 30% volatility and correlation 0.2. With $a = (1, 1)$ the result is portfolio C: weights 73.6% / 26.4%, volatility 18.1%, and both assets have beta exactly 1 to it (2A.16).
- *Use:* turns any signal into its purest, minimum-risk long/short portfolio. This is the bridge to information analysis in Ch. 12.

5. Maximum Sharpe ratio and the correlation rule (2A.24), (2A.28)

```
\mathrm{SR}_q = \sqrt{f^{\top} V^{-1} f}, \qquad \mathrm{SR}_P = \rho_{P,q}\,
\mathrm{SR}_q
```

- *Derivation:* scaling a portfolio doesn't change its Sharpe ratio, so fix expected return at 1 and minimize variance; that is the characteristic portfolio of f. The second identity follows by multiplying $f = Vh_q / \sigma_q^2$ by h_P and dividing by σ_P.

- *Intuition:* any portfolio's Sharpe ratio is the best Sharpe ratio times how correlated it is with the best portfolio.
- *Example:* expected excess returns 5% and 8% on the two assets above give SR_q = 0.334. A 50/50 portfolio has correlation 0.992 with Q and SR = 0.331, as the identity predicts.
- *Use:* explains why a slightly imperfect portfolio loses little Sharpe, and why a portfolio that is uncorrelated with your alpha portfolio earns nothing.

6. Reverse optimization: returns explained by Q (2A.36)

```
f = f_Q \cdot \frac{V h_Q}{\sigma_Q^2} = f_Q\, \beta_{\cdot,Q}
```

- *Intuition:* if Q is optimal, each asset's expected return is proportional to its beta with Q. Setting Q = M is the CAPM; setting Q = B gives "consensus" for any benchmark.
- *Example:* a 60/40 benchmark of the two assets above with f_B = 6% implies expected excess returns of 5.0% and 7.5% (betas 0.83 and 1.25). Feed those back in and the optimizer returns 60/40.
- *Use:* the neutral starting point for alpha generation and for Black–Litterman-style blending (outside the book).

7. The efficient frontier (2A.45), (2A.47)–(2A.48)

```
h_P = w\, h_C + (1-w)\, h_Q, \qquad \sigma_P^2 = \sigma_C^2 + \kappa\,(f_P - f_C)^2, \qquad \kappa = \frac{\sigma_Q^2 - \sigma_C^2}{(f_Q - f_C)^2}
```

- *Example (Fig. 2A.1, p. 37):* σ_Q = 20%, f_Q = 7%, σ_C = 12% imply f_C = 2.52% (from $f_C/\sigma_C^2 = f_Q/\sigma_Q^2$) and κ = 12.76. The fully invested efficient portfolio with 5% expected excess return has 14.9% risk.
- *Use:* two-fund separation; every efficient fully invested portfolio mixes the minimum-variance and maximum-Sharpe portfolios.

D. Trading implications

1. **Neutral prior [book]:** reverse-optimize the benchmark and express views only as deviations from it.
2. **Beta-neutral research [book]:** with β = 1 you need no forecast of μ_M at all (p. 24); spend research on residuals.
3. **Signals as portfolios [book, appendix]:** evaluate a signal through its characteristic portfolio's return and risk, not through raw correlations.

4. **Beta shrinkage [book, p. 14 fn. 4]:** historical betas mean-revert; shrink them toward 1 or use fundamental betas.
5. **Low-beta hypothesis [outside the book, needs verification]:** if the empirical security market line is flatter than the CAPM line (Black 1972 restricted borrowing, cited on p. 25; later Frazzini and Pedersen 2014), the gap between consensus and realized returns is itself a candidate alpha.

E. Practical implementation

```
import numpy as np, gk_part1 as gk

def shrunk_beta(r_p, r_m, prior=1.0, prior_se=0.3):
    """Bayesian (Vasicek-style) shrinkage of a historical beta toward a
prior."""
    fit = gk.ols_alpha_beta(r_p, r_m)
    w = prior_se**2 / (prior_se**2 + fit['se_beta']**2)
    return w * fit['beta'] + (1 - w) * prior

def consensus_returns(V, h_B, f_B=0.06):
    return gk.implied_excess_returns(V, h_B, f_B)          # (2A.36) with Q =
B

def signal_portfolio(V, signal):
    h, var = gk.characteristic_portfolio(V, signal)        # (2A.1): min-
risk, unit exposure
    return h, np.sqrt(var)
```

The toolkit's tests verify (2A.3), (2A.4), (2A.16), (2A.19), (2A.28), (2A.36) and (2A.47) on random covariance matrices.

F. Critical evaluation

- **Empirical CAPM:** the authors acknowledge Fama and French (1992) and defer to Black (1993) and Grinold (1993) (p. 22). Since then, the flat security market line and multifactor pricing are mainstream; consensus returns are better built from a factor model than from beta alone.
- **Unobservable market:** the CAPM needs the market of all assets. Replacing it with an equity benchmark (Ch. 4) makes "consensus" a convention, not an equilibrium.

- **The zero-sum argument** is exact for cap-weighted residuals, but it says nothing about who pays. Liquidity providers, investors with tax or regulatory constraints, and short-horizon traders can rationally accept negative residual returns.
- **Noise:** the American Express example shows one stock's alpha over 60 months has a standard error of about 11% a year (96 bp monthly × $\sqrt{12}$). Realized alphas are mostly noise.
- **Covariance dependence:** every appendix result uses V^{-1}. Reverse-optimized returns inherit every error in V; Ch. 3 is the remedy.

G. Key takeaways

1. Beta separation is free; the CAPM's only claim is that residuals have zero mean.
2. Consensus expected returns are those that make the benchmark optimal; active management is the deviation.
3. Any attribute maps one-to-one to a minimum-risk portfolio, so signals can be studied as portfolios.
4. $SR_P = \rho(P, Q) \times SR_Q$: correlation with the ideal portfolio is everything.
5. Historical average returns are nearly useless as forecasts ($\sigma/\sqrt{Y}$).

Test your understanding

Conceptual questions

1. Why does the beta decomposition need no theory while the CAPM does?
2. Explain the investor A/B/C argument and one real-world reason it might not imply zero expected residual returns.
3. Why does an optimizer fed CAPM returns produce the market plus cash, and what changes if the minimum-variance portfolio has a negative expected return?
4. What is the economic meaning of a characteristic portfolio for book-to-price?
5. Why does a beta-neutral manager not need a forecast of the market premium?

Math exercises

1. Two stocks: $\beta_A = 1.15$, $\sigma_A = 35\%$; $\beta_B = 0.95$, $\sigma_B = 33\%$; $\sigma_M = 20\%$; residuals uncorrelated. Find their correlation and which has more residual risk.
2. What expected excess returns would make a 100% position in GE optimal (Problem 4, p. 24)?
3. Prove that $\beta_C = \sigma_C^2 / \sigma_M^2 \leqslant 1$ (Exercise 1, p. 38).

Solutions. (1) $\rho = 1.15 \cdot 0.95 \cdot 400 / (35 \cdot 33) = 0.378$; $\omega_A = 26.4\%$, $\omega_B = 27.0\%$, so B has more. (2) Any f proportional to V e_{GE}, i.e. $f_n = \kappa \operatorname{Cov}(r_n, r_{GE})$ with $\kappa > 0$: every stock's expected return is proportional to its beta with GE (2A.36 with Q = GE). (3) From (2A.17) every fully invested P has $\operatorname{Cov}(r_P, r_C) = \sigma_C^2$; with P = M, $\beta_C = \operatorname{Cov}(r_C, r_M)/\sigma_M^2 = \sigma_C^2/\sigma_M^2$, and $\sigma_C \leqslant \sigma_M$ because C is the minimum-variance fully invested portfolio.

Coding exercises

1. Simulate 60 months of returns for 500 stocks with true betas drawn from $N(1, 0.3^2)$. Compare the out-of-sample error of raw OLS betas and shrunk betas in the next 60 months.
2. Using `characteristic_portfolio`, build the minimum-risk portfolio with unit exposure to a value score on a factor-model covariance. Report its gross leverage and how leverage scales if you multiply the score by 10 (Prop. 1 item 5).

Research experiment

On 20+ years of point-in-time US equity data, sort stocks into beta deciles using shrunk pre-ranking betas. Compare realized excess returns with CAPM-implied ones (β × realized market premium), net of a realistic cost model. The slope of the realized line versus the CAPM slope tests whether a low-beta tilt is a usable alpha in your universe.

Connections: Ch. 1 promised that every characteristic is a portfolio; this chapter delivers the algebra. Chapter 3 supplies the V that every formula here needs, and Ch. 4 replaces the market with a benchmark.

7. Chapter 3 — Risk (pp. 41–86)

A. Central thesis

Risk is the annual standard deviation of return, its cost is proportional to variance, and active managers care mostly about active and residual risk. Because a raw covariance matrix cannot be estimated reliably, risk should come from a structural multifactor model, which also explains where the risk comes from (p. 70).

B. Key concepts

- **Criteria for a risk measure (pp. 41–42):** universal, symmetric (missing a winner hurts as much as holding a loser), flexible across assets and horizons, and accurately forecastable.

- **Alternatives rejected (pp. 44–46):** semivariance, target semivariance, shortfall probability and VaR are ambiguous, hard to aggregate from assets to portfolios, and hard to forecast. Magellan's semivariance was 55% of its variance, close to the symmetric 50%.
- **Risks don't add (p. 47):** portfolio risk is below the weighted average of asset risks unless correlations are 1. This is the source of diversification.
- **Time scaling (p. 49):** with no autocorrelation, variance grows linearly with horizon and risk with its square root.
- **Active risk (tracking error) and residual risk (pp. 49–51):** active risk depends on the size of the active position and the stock's risk, not on its benchmark weight (p. 50). Typical large-cap stocks have 25–40% total risk, 20–35% residual risk, and betas of 0.8–1.35 (Table 3.1).
- **Elementary models (pp. 52–54):** single-factor, constant-correlation, and historical covariance. The last is "neither robust nor reasonable": with T ⩽ N some active positions look riskless.
- **Structural models (pp. 55–63):** returns = exposures × factor returns + specific return. Factors must be known a priori and fall into three families: macro responses, cross-sectional attributes, statistical factors. The authors favour cross-sectional factors (industries and risk indices such as volatility, momentum, size, liquidity, growth, value, earnings volatility, leverage).
- **Uses (pp. 64–67):** present (decompose risk; classify bets as inherent, intentional or incidental), future (portfolio construction), past (performance attribution).
- **Evidence (pp. 67–70):** in Kahn (1997), structural forecasts for 29 funds produced one two-sigma surprise (3%) against five (17%) for historical forecasts. In Kahn and Stefek (1996), risk measures other than variance showed no persistence beyond what variance implies.
- **Appendix (pp. 72–84):** GLS cross-sectional regressions, factor portfolios, specific-risk modelling, marginal contributions, risk attribution, and why correlations rise with market volatility.

C. Mathematical foundations

1. Two-asset risk and diversification (3.1)–(3.2)

```
\sigma_P = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2 w_1 w_2 \sigma_1
\sigma_2 \rho_{12}} \;\le\; w_1\sigma_1 + w_2\sigma_2
```

- *Example (Fig. 3.2):* GE 27.4%, IBM 29.7%, correlation 0.629. The minimum-risk mix is 60.8% GE with 25.6% risk, below either stock.
- *Derivation of the minimum:* setting $d\sigma^2/dw = 0$ gives $w_1 = (\sigma_2^2 - \rho\sigma_1\sigma_2) / (\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$.

2. Equal-correlation diversification (3.4)–(3.5)

```
\sigma_P = \sigma\sqrt{\frac{1 + \rho\,(N-1)}{N}}
\;\xrightarrow{N\to\infty}\; \sigma\sqrt{\rho}
```

- *Derivation:* with equal weights 1/N, variance = $N(1/N^2)\sigma^2 + N(N-1)(1/N^2)\rho\sigma^2$.
- *Example (p. 49):* the 20 MMI stocks average 27.8% risk and the equal-weighted portfolio has 20.4%; inverting gives an average correlation of about 0.51 (the book rounds to 0.52).
- *Use:* a fast sanity check on any risk number; also shows why factor (correlated) risk cannot be diversified away.

3. Time scaling (3.6)

```
\sigma_{\text{annual}} = \sqrt{12}\;\sigma_{\text{monthly}}
```

- *Example (p. 49):* 5% annual active risk is 2.5% over a quarter and 10% over four years.
- *Assumption:* no autocorrelation. Smoothed or illiquid returns break it and understate risk.

4. Active and residual risk, and the cost of risk (3.7)–(3.9)

```
\psi_P = \operatorname{Std}\{r_P - r_B\}, \qquad \omega_P = \sqrt{\sigma_P^2
- \beta_P^2 \sigma_B^2}, \qquad \text{cost} = \lambda\,\psi_P^2
```

- *Example (p. 100 and Ch. 4 Problem 3):* $\sigma_B = 20\%$, $\sigma_P = 21\%$, $\beta = 1$ gives $\omega = 6.4\%$. One point of total risk hides 6.4 points of residual risk.
- *Example (p. 52):* with $\lambda = 0.1$, 4% active risk costs $0.1 \times 16 = 1.6\%$ of expected return.

5. The structural risk model (3.16)–(3.17), (3A.1)–(3A.2)

```
r = X b + u, \qquad V = X F X^{\top} + \Delta
```

- *Variables:* X is N × K exposures known at the start of the period; b are factor returns; u are specific returns; F is the factor covariance; Δ is diagonal specific variance.
- *Assumptions:* $\text{Cov}(u, b) = 0$ and $\text{Cov}(u_n, u_m) = 0$ for $n \neq m$ (A1–A2, p. 72).

- *Why it works:* for 6,000 stocks a raw covariance has 18,003,000 free numbers. A 68-factor model needs 2,346 factor covariances plus 6,000 specific variances, about 8,346 (p. 55).
- *Standardization (3.18):* risk-index exposures are z-scored across the universe, e.g. GE's size exposure of 1.90 in Sep 1998 (p. 63).

6. Factor returns by GLS and factor portfolios (3A.3)–(3A.4)

```
b = (X^{\top}\Delta^{-1}X)^{-1} X^{\top}\Delta^{-1} r, \qquad b_k = \sum_n
c_{k,n}\, r_n
```

- *Intuition:* each factor return is the return of a portfolio with unit exposure to that factor, zero to all others, and minimum specific risk.
- *Practice (p. 73):* about 1,500 stocks and 65 factors, one regression a month, R^2 of 30–40%. Weighting by $\sqrt{\text{cap}}$ mimics inverse specific variance; specific variance shrinks by a factor of about 0.7 each time company size doubles (fn. 11).

7. Specific risk (3A.5)

```
u_n^2(t) = S(t)\,[1 + v_n(t)]
```

S(t) is the average specific variance (a time-series model) and v_n(t) the cross-sectional deviation (a factor model on risk indices and recent squared residuals).

8. Marginal contributions and risk attribution (3A.20)–(3A.23), (3A.31)–(3A.32)

```
\mathrm{MCAR} = \frac{V h_{PA}}{\psi_P}, \qquad h_{PA}^{\top}\,\mathrm{MCAR}
= \psi_P, \qquad \text{share}_n = \frac{h_{PA}(n)\,\mathrm{MCAR}(n)}{\psi_P}
```

- *Derivation:* $\psi = \sqrt{(h^{\mathsf{T}}Vh)}$ is homogeneous of degree 1 in h, so Euler's theorem gives $h^{\mathsf{T}}\,\partial\psi/\partial h = \psi$. Each pairwise covariance is split evenly between its two assets (p. 82 fn. 15).
- *Example (three assets, toolkit run):* volatilities 20/30/25%, correlations 0.2/0.3/0.5, benchmark 50/30/20, portfolio 60/20/20. Tracking error is 3.26%. MCAR is (0.086, −0.240, −0.069): adding 1% to asset 1 (from cash) raises tracking error by about 8.6 bp. Attribution is 26% asset 1, 74% asset 2, 0% asset 3, even though asset 3 has a nonzero marginal contribution.
- *Book example (Table 3.3):* in the MMI vs S&P 500, IBM had the largest MCAR (0.074) and AT&T the only negative one (−0.009).

9. Correlations rise with market volatility (3A.40)–(3A.41)

```
\rho_{nm} \to \frac{\sigma_M^2}{\sigma_M^2 + \omega^2}
```

- *Example:* two β = 1 stocks with 25% residual risk are 0.26 correlated when the market runs at 15% and 0.59 when it runs at 30%.
- *Use:* diversification is weakest exactly when markets are volatile. Stress-test portfolios at high market volatility.

10. Fast inversion (Exercise 5, p. 84)

```
V^{-1} = \Delta^{-1} - \Delta^{-1}X\,(X^{\top}\Delta^{-1}X + F^{-1})^{-1}
X^{\top}\Delta^{-1}
```

Only K × K matrices are inverted, which makes N = 5,000 optimizations cheap.

D. Trading implications

1. **Set limits in risk, not in benchmark-relative percentages [book, p. 50].** A 1% active position in a small stock is not riskier than 1% in a large one if their risks match.
2. **Hunt incidental bets [book, p. 65].** Decompose active risk by factor; anything you did not intend to bet on should be neutralized.
3. **Use structured covariance in optimizers [book; E4].** A sample covariance looked fine for random portfolios but optimized portfolios realized 5.6× their predicted risk.
4. **Check the alpha–risk link [book, p. 65].** The highest-alpha names should carry the highest marginal contributions to risk.
5. **Plan for correlation spikes [book appendix + inference].** Size gross exposure for the high-volatility regime, not the average one.

E. Practical implementation

```
import numpy as np, gk_part1 as gk

def estimate_factor_model(R, X, halflife=None):
    """R: T x N excess returns; X: N x K exposures (held fixed here for
brevity)."""
    T, N = R.shape
    b, u = zip(*(gk.gls_factor_returns(R[t], X, np.ones(N)) for t in
range(T)))   # pass 1: OLS
    d = np.var(np.array(u), axis=0, ddof=1)
    b, u = zip(*(gk.gls_factor_returns(R[t], X, 1/d) for t in range(T)))
# pass 2: GLS
```

```
    b, u = np.array(b), np.array(u)
    w = np.ones(T) if halflife is None else 0.5 ** (np.arange(T)[::-1] /
halflife)
    w /= w.sum()
    bc = b - w @ b
    F = (bc * w[:, None]).T @ bc
# EWMA factor covariance
    return gk.FactorRiskModel(X, F, np.var(u, axis=0, ddof=1))

def risk_report(model, h_P, h_B):
    h_PA = h_P - h_B
    d = model.risk_decomposition(h_PA)
    V = model.cov()
    return dict(tracking_error=d['total'], factor_share=d['factor_var'] /
d['total']**2,
                asset_attribution=gk.active_risk_attribution(h_P, h_B, V),
                factor_mcar=model.factor_marginal_contribution_active(h_PA))
```

Experiment E4 in the toolkit runs this estimator on simulated data and compares predicted with realized risk.

F. Critical evaluation

- **Tails are waved away too quickly.** The Kahn and Stefek result concerns persistence of abnormal tail measures across funds, not whether portfolio tails matter. Crowded factor portfolios, such as the August 2007 quant unwind (outside the book), suffer losses far beyond a normal-model forecast.
- **"Active returns should be symmetric by construction" (p. 45)** fails for option-like strategies and for strategies short volatility or liquidity.
- **Thin evidence.** The model-validation study used 29 funds over one year (1993–94). It supports the claim but cannot prove it.
- **Statistical factors are dismissed** as uninterpretable and unable to capture momentum (p. 59). Modern practice uses them as a supplement for short horizons and for catching factors the fundamental model misses.
- **Vendor perspective.** The authors built BARRA models (p. 55 fn. 8); a real risk model still has to be validated against your own portfolios with bias statistics.
- **Alpha–risk misalignment** (outside the book): if your alpha uses factors the risk model lacks, the optimizer treats them as diversifiable specific risk and overloads them.

G. Key takeaways

1. Standard deviation wins because it aggregates and forecasts; report other measures if you like, but optimize on variance.
2. Risks don't add, and correlated risk has a floor of $\sigma\sqrt{\rho}$.
3. Active risk depends on active weight × stock risk, not on benchmark weight.
4. Structural factor models turn an impossible estimation problem into a tractable one.
5. Marginal contributions and Euler attribution tell you where risk lives and whether it matches your alphas.

Test your understanding

Conceptual questions

1. Why do the authors require factor exposures to be known a priori, and which factor types struggle with that?
2. Why is a historical covariance matrix with $T \leqslant N$ dangerous inside an optimizer but harmless for measuring an index fund's risk?
3. Distinguish inherent, intentional and incidental bets with an example from your own portfolio.
4. Why are asset correlations higher in volatile markets under a single-factor model?
5. When would semivariance carry information that variance does not?

Math exercises

1. Stock A has 25% risk, stock B 50%, correlation 0.5. Find the minimum-risk fully invested mix (Problem 2, p. 70).
2. An equal-weighted portfolio of 50 stocks with 25% residual risk each tracks an equal-weighted index under the single-factor model. Estimate the tracking error (Exercise 4, p. 84).
3. Show that $h_B^T \, MCRR = 0$ (Exercise 3, p. 84) and interpret it.

Solutions. (1) $w_A = (2500 - 625) / (625 + 2500 - 1250) = 1.0$: hold 100% A at 25% risk; adding B never lowers risk because $\rho\sigma_B/\sigma_A = 1$. (2) $\omega \approx \omega_0/\sqrt{N} = 25\%/\sqrt{50} = 3.5\%$. (3) $MCRR = V h_{PR} / \omega$ and $h_B^T V h_{PR} = \sigma_B^2 \beta^T h_{PR} = 0$ because residual holdings have zero beta. Adding benchmark to a portfolio changes only its beta, never its residual risk.

Coding exercises

1. Implement the bias statistic: for each month, standardized active return z = r_A / ψ_pred; report the rolling 12-month standard deviation of z and the share of months outside ±2.
2. Extend `risk_report` to group industry factors into sectors using portfolio industry weights (p. 81) and compare with benchmark weights.

Research experiment

On real daily data for the Russell 1000, build (a) a sample covariance, (b) a Ledoit–Wolf shrinkage covariance, (c) a 10-factor statistical model, and (d) a fundamental factor model. For each, form monthly minimum-tracking-error portfolios of 50 names against the index and record bias statistics out of sample. The hypothesis from Ch. 3 and E4 is that (a) underpredicts risk badly for optimized portfolios and (d) is close to unbiased.

Connections: Ch. 2's formulas all need V^{-1}; this chapter supplies V. The marginal contributions here become the optimality condition α = IR × MCRR in Ch. 5, and the factor attribution reappears in performance analysis (Ch. 17).

8. Chapter 4 — Exceptional Return, Benchmarks, and Value Added (pp. 87–108)

A. Central thesis

Replace "the market" with the benchmark the client actually assigns, split every expected return into four parts, and price benchmark risk and active risk differently. The client bears benchmark risk; the manager bears active risk, so the manager's objective is value added = (timing return − timing risk penalty) + (alpha − residual risk penalty) (4.15).

B. Key concepts

- **Benchmark (pp. 88–89):** the sponsor's instruction to a specialist manager ("bogey", "normal portfolio"). It enables specialization and focuses attention on relative performance.
- **Relative beta and active positions (p. 90):** beta, residual return and active holdings are all defined against the benchmark. Active cash is minus the sum of active risky holdings (4.3).
- **Four components of expected return (pp. 90–91):** time premium i_F, risk premium βμ_B, exceptional benchmark return βΔf_B, and alpha.

- **Three combinations (p. 92):** consensus expected excess return βμ_B; expected excess return βf_B + α; exceptional expected return βΔf_B + α, which is where all active management lives.
- **Total risk/return management (pp. 93–99):** with one risk aversion, tiny bits of information produce huge residual risk (Table 4.3: ω = 13.8% for MMI stocks with historical alphas).
- **Business risk versus investment risk (pp. 100–101):** the tide of benchmark return lifts all managers equally; residual return decides who is fired. Hence a much higher aversion to residual risk.
- **Benchmark timing (pp. 101–102):** a separate decision, set aside until Ch. 19 for four reasons, including that it rarely adds much value (fundamental law).
- **Active vs residual (pp. 102–103):** with β = 1 they are identical; with timing, active return = residual return + active beta × benchmark return.

C. Mathematical foundations

1. Active variance (4.4)–(4.5), (4.20)

```
\psi_P^2 = h_{PA}^{\top} V h_{PA} = \beta_{PA}^2\,\sigma_B^2 + \omega_P^2
```

- *Derivation:* r_PA = θ_P + β_PA r_B (4.19); θ_P is uncorrelated with r_B, so the variances add.
- *Example:* β_P = 1.05, σ_B = 20%, ω_P = 4% gives ψ = √(1 + 16) = 4.12%. A 0.05 beta drift adds little tracking error at 4% residual risk, but it is an unpaid bet unless you meant it.

2. Decomposition of expected return (4.7)–(4.9)

```
E\{R_n\} = 1 + i_F + \beta_n \mu_B + \beta_n \Delta f_B + \alpha_n, \qquad
\Delta f_B = f_B - \mu_B
```

- *Variables:* i_F is the risk-free rate; μ_B the long-run (consensus) benchmark excess return, typically 3–7% a year (p. 91); f_B your near-term forecast; α_n the expected residual return.
- *Example (Table 4.1, p. 92):* MMI in Dec 1992: risk-free 0.26%, risk premium 0.96 × 0.50 = 0.48%, benchmark surprise 1.31 − 0.26 − 0.50 = 0.55% so exceptional benchmark return 0.96 × 0.55 = 0.53%, and alpha −1.87%, summing to the realized −0.60%.
- *Example (Problem 1, p. 104):* McDonald's, β = 1.07, expected total 15%, i_F = 6%, f_B = 6.5%, μ_B = 6%: risk premium 6.42%, exceptional benchmark 0.535%, alpha 2.045%; consensus 12.42% + exceptional 2.58% = 15%.

- *Use:* a spreadsheet that strips analysts' total-return targets into the only part that should move the portfolio (the alpha).

3. The risk aversion that makes the benchmark optimal (4.12)

```
\lambda_T = \frac{\mu_B}{2\,\sigma_B^2}
```

- *Derivation (p. 97 fn. 4):* mixing B with cash, maximize $\beta\mu_B - \lambda_T\beta^2\sigma_B^2$. The first-order condition $\mu_B = 2\lambda_T\beta\sigma_B^2$ gives $\beta = 1$ exactly when (4.12) holds.
- *Example:* $\mu_B = 6\%$, $\sigma_B = 20\%$ gives 0.0075; $\mu_B = 8\%$, $\sigma_B = 16\%$ gives 0.0156.

4. Why one risk aversion is too aggressive (4.13)–(4.14) and p. 99 fn. 6

```
\beta_P = 1 + \frac{\Delta f_B}{\mu_B}, \qquad \omega_P = \frac{\mathrm{IR}}
{2\,\lambda_T}
```

- *Example:* a modest view that the benchmark returns 7.5% instead of 6% sets $\beta_P = 1.25$. A manager with IR = 0.75 and $\lambda_T = 0.01$ would run 37.5% residual risk. Real equity managers run about 4–5% (Table 5.8).
- *Intuition:* total risk is dominated by the benchmark, so the residual bet looks almost free in total-risk terms. The client does not see it that way.

5. The value-added objective (4.15), derived as (4A.4)–(4A.8)

```
\mathrm{VA} = \underbrace{\beta_{PA}\Delta f_B -
\lambda_{BT}\,\beta_{PA}^2\sigma_B^2}_{\text{benchmark timing}} \;+\;
\underbrace{\alpha_P - \lambda_R\,\omega_P^2}_{\text{stock selection}}
```

- *Derivation:* write $f_P = f_B + \beta_{PA}\mu_B + \beta_{PA}\Delta f_B + \alpha_P$ and $\sigma_P^2 = \sigma_B^2 + 2\beta_{PA}\sigma_B^2 + \beta_{PA}^2\sigma_B^2 + \omega_P^2$. Pair the terms: $(f_B - \lambda_T\sigma_B^2)$ is a constant; $\beta_{PA}(\mu_B - 2\lambda_T\sigma_B^2)$ is zero by (4.12); the remaining pairs get their own aversions λ_{BT} and λ_R.
- *Sanity check (p. 103):* with a cash benchmark and $\lambda_T = \lambda_{BT} = \lambda_R$ you recover ordinary mean/variance.

6. Optimal active beta (4.16)

```
\beta_{PA}^* = \frac{\Delta f_B}{2\,\lambda_{BT}\,\sigma_B^2}
```

- *Example:* $\Delta f_B = 1\%$, $\sigma_B = 20\%$. With $\lambda_{BT} = 0.075$ the active beta is 0.017; with $\lambda_{BT} = \lambda_T = 0.0075$ it is 0.167, ten times larger.
- *Use:* if you have no timing view ($\Delta f_B = 0$) or very high timing aversion, beta stays at 1.

D. Trading implications

1. **Feed the optimizer exceptional returns only [book].** Analysts' total-return targets must be stripped of time premium, risk premium and any timing view first.
2. **Calibrate residual aversion from a tracking-error budget [book, (5.11)].** Don't use a textbook utility parameter.
3. **Separate timing from selection [book].** Run them as two books with separate budgets, or switch timing off.
4. **Don't use historical alphas as forecasts [book, Table 4.2].** Coca-Cola's 20% historical alpha and IBM's −19% are noise that would dominate any optimizer.
5. **Watch benchmark misfit [inference].** If your benchmark is not efficient (e.g. an S&P 500 mandate run with a small-cap tilt), part of your "alpha" is factor premium the client could buy cheaply.

E. Practical implementation

```
import gk_part1 as gk

d = gk.decompose_expected_return(total_return=0.15, beta=1.07, i_F=0.06,
mu_B=0.06, f_B=0.065)
# {'risk_premium': 0.0642, 'exceptional_benchmark': 0.00535, 'alpha':
0.02045, ...}

def value_added_optimal(dfB, sigma_B, lam_BT, IR, lam_R):
    b = gk.optimal_active_beta(dfB, lam_BT, sigma_B)                 # (4.16)
    w = gk.optimal_residual_risk(IR, lam_R)                           #
(5.10)
    return dict(active_beta=b, residual_risk=w,
                VA=gk.value_added(b, dfB, lam_BT, sigma_B, IR * w, lam_R, w))
# (4.15)
```

F. Critical evaluation

- **Separate aversions are an institutional patch, not utility theory.** The authors say so (p. 100). Roll (1992), cited on p. 105, shows that minimizing tracking error for a given alpha produces portfolios that are inefficient in total-risk space; the sponsor bears that cost.
- **μ_B is a guess** ("a number between 3 and 7 percent", p. 91). Luckily, a beta-neutral manager does not need it.

- **Benchmark choice is explicitly out of scope** (p. 88). Benchmark misfit is one of the largest sources of mismeasured skill in practice.
- **Absolute-return capital** (hedge funds, prop trading) does not fit the business-risk story; there the cash benchmark version applies and total risk and drawdowns matter.

G. Key takeaways

1. Expected return = time premium + risk premium + exceptional benchmark return + alpha.
2. Only the exceptional parts carry information; consensus inputs lead back to the benchmark.
3. Total-risk utility makes residual bets look cheap and produces absurd residual risk.
4. Value added = timing term + selection term, each with its own risk aversion.
5. With $\beta = 1$, active risk equals residual risk, and the rest of the book can focus on alpha versus ω.

Test your understanding

Conceptual questions

1. Why does a mean/variance investor with total-risk aversion take enormous residual risk on small alphas?
2. Why is the cross-effect term $\beta_{PA}(\mu_B - 2\lambda_T\sigma_B^2)$ zero, and what would it mean if it weren't?
3. Under the CAPM with a non-market benchmark, how do expected returns split into the four components (Problem 2)?
4. Who bears benchmark risk and who bears active risk in a pension plan?
5. Why does a no-timing manager not need an estimate of μ_B?

Math exercises

1. $\sigma_B = 20\%$, $\sigma_P = 21\%$, $\beta = 1$. Find residual risk, active risk, and compare with $\sigma_P - \sigma_B$ (Problem 3).
2. Investor A maximizes $f - 0.0075\sigma^2$; investor B maximizes $\alpha - 0.075\omega^2$. Portfolio 1: $f = 10\%$, $\sigma = 20.22\%$; portfolio 2: $f = 16\%$, $\sigma = 25\%$; both $\beta = 1$, $f_B = 6\%$, $\sigma_B = 20\%$. Who picks what (Problem 4)?
3. With $\lambda_T = 0.0075$, what certainty-equivalent return does a portfolio with $f = 6\%$ and $\sigma = 20\%$ offer (Problem 5)?

Solutions. (1) $\omega = \psi = \sqrt{}(441 - 400) = 6.40\%$ versus a 1-point difference in total risk. (2) A: $U_1 = 10 - 3.07 = 6.93$, $U_2 = 16 - 4.69 = 11.31$, so A picks portfolio 2. B: $\alpha_1 = 4$, $\omega_1 = 2.97\%$, $VA_1 = 3.34$; $\alpha_2 = 10$, $\omega_2 = 15\%$, $VA_2 = -6.88$, so B picks portfolio 1. (3) $6 - 0.0075 \times 400 = 3\%$.

Coding exercises

1. Build the Table 4.2 spreadsheet in pandas: given total-return forecasts, betas, i_F, μ_B and f_B, output the four components and flag names whose alpha exceeds 2× a plausible bound (IC × ω × 3).
2. Solve max VA over (β_PA, h_PR) numerically with cvxpy and confirm it matches (4.16) and (5.10).

Research experiment

Take sell-side 12-month price targets for a large universe. Convert them to implied total returns, strip time premium and βμ_B as in (4.7), and measure the IC of the resulting raw "alphas" with next-12-month residual returns. Then check how much of the IC survives after neutralizing industry and size. The hypothesis is that most raw target-implied alpha is beta, industry or size, not stock-specific information.

Connections: Ch. 2's consensus returns become βμ_B here, and Ch. 3's active and residual risk become the penalty terms. Chapter 5 takes the selection half of (4.15) and asks how much value it can produce.

9. Chapter 5 — Residual Risk and Return: the Information Ratio (pp. 109–146)

A. Central thesis

The information ratio, the best attainable ratio of expected residual return to residual risk, is the active manager's budget constraint. Combined with the value-added objective it fixes the optimal aggressiveness, $\omega^* = IR / 2\lambda$, and the achievable value added, $IR^2 / 4\lambda$. Every investor, whatever their risk aversion, prefers the strategy with the highest IR (p. 125).

B. Key concepts

- **Alpha (pp. 111–112):** ex ante, a forecast of residual return; ex post, the average realized residual. Alphas have the portfolio property and must be benchmark-neutral ($\alpha_B = 0$).

- **Ex-post IR (p. 112):** realized residual return over residual risk; roughly the alpha t-statistic divided by √(years).
- **Ex-ante IR (pp. 113–115):** "the assumed can opener". A start-up manager hoping for 3.5% alpha at 5.5% risk has IR ≈ 0.64. The generic before-fee distribution is symmetric around zero, with 0.5 at the top quartile and 1.0 at the top decile (Table 5.1).
- **IR is independent of aggressiveness (p. 116):** scaling active positions scales alpha and risk equally (Table 5.2, portfolios P and L), until constraints bind.
- **IR depends on horizon (p. 117):** it grows with √T, so a monthly IR is 0.29× the annual IR.
- **Residual frontier (pp. 117–119):** a straight line α = IR × ω through the benchmark at the origin.
- **Preferences meet opportunities (pp. 119–125):** value-added curves are parabolas; the optimum is the tangency point.
- **β = 1 and no-active-cash frontiers (pp. 125–127):** each extra constraint shrinks the opportunity set.
- **Forecast alphas directly (pp. 127–129):** rank into five bins at +2/+1/0/−1/−2%, then subtract benchmark-average alpha × beta.
- **Empirics (pp. 129–132):** top-quartile equity IRs of 0.63–0.78 before fees and 0.48–0.58 after; median equity active risk 4.4–4.8%; median bond active risk 0.6–1.3% (Tables 5.6–5.9).
- **Appendix (pp. 134–145):** portfolio A, ten properties linking A, Q and the IR, the optimal policy, portfolio Y (no active beta or cash), time scaling, and linearity of IR in the alphas.

C. Mathematical foundations

1. Definition and the ex-post link to t-statistics (5.5)–(5.6)

$$\mathrm{IR}_P = \frac{\alpha_P}{\omega_P}, \qquad \mathrm{IR} = \max_P \mathrm{IR}_P, \qquad t_\alpha \approx \mathrm{IR}\sqrt{Y}$$

- *Example (p. 112):* 2.3% residual return at 3.45% risk is IR = 0.67. After 5 years that is t ≈ 1.5, not significant; reaching t = 2 takes about 9 years.

2. The attainable IR from a vector of alphas (5A.1)–(5A.7)

$$h_A = \frac{V^{-1}\alpha}{\alpha^{\top}V^{-1}\alpha}, \qquad \mathrm{IR} = \sqrt{\alpha^{\top} V^{-1} \alpha}, \qquad \omega_A = \frac{1}{\mathrm{IR}}$$

- *Derivation:* the IR is scale-free, so fix $\alpha_P = 1$ and minimize risk; that is the characteristic portfolio of α (Ch. 2). Its variance is $1/(\alpha^T V^{-1}\alpha)$, so its IR is the square root of $\alpha^T V^{-1}\alpha$. Benchmark-neutral alphas make h_A zero-beta (Prop. 1.1).
- *Example:* two uncorrelated stocks with alphas +2% and −1% and residual risks 25% and 30% give IR = $\sqrt{(0.08^2 + 0.03^3)} = 0.087$. With 500 independent stocks each at $\alpha/\omega = 0.04$, IR = $0.04\sqrt{500} = 0.89$. Squared IRs add across independent assets, which previews Ch. 6.
- *Use:* compute the IR your alphas imply before optimizing; if it is 2.5, your alphas are overconfident.

3. Residual frontier and value added (5.7)–(5.9)

```
\alpha_P = \mathrm{IR}\cdot\omega_P, \qquad \mathrm{VA}[P] = \alpha_P -
\lambda_R\,\omega_P^2, \qquad \mathrm{VA}[\omega] = \omega\,\mathrm{IR} -
\lambda_R\,\omega^2
```

- *Example (Fig. 5.4):* with $\lambda_R = 0.10$, the point ($\alpha = 4.1\%$, $\omega = 4\%$) has the same value added, 2.5%, as a riskless 2.5% alpha.

4. Optimal aggressiveness and optimal value added (5.10)–(5.12)

```
\omega^* = \frac{\mathrm{IR}}{2\lambda_R}, \qquad \lambda_R =
\frac{\mathrm{IR}}{2\,\omega^*}, \qquad \mathrm{VA}^* = \frac{\mathrm{IR}^2}
{4\lambda_R} = \frac{\omega^*\,\mathrm{IR}}{2}
```

- *Derivation:* $d/d\omega\ (\omega\,IR - \lambda\omega^2) = IR - 2\lambda\omega = 0$. Substituting back gives $IR^2/2\lambda - IR^2/4\lambda$.
- *Grid (Tables 5.3–5.4):* IR 0.5/0.75/1.0 against λ 0.05/0.10/0.15 spans ω^* from 1.67% to 10% and VA* from 0.42% to 5.00%.
- *Example (pp. 124–125):* the start-up manager (IR 0.64, ω 5.5%) implies $\lambda = 0.058$ and VA* = 1.77% a year.
- *Overconfidence penalty (derived here from (5.9)):* if you size risk for IR_assumed = k × IR_true, value added is $(2k - k^2) \times VA^*$. Overstating IR by 50% keeps 75% of value added; overstating it by 2× leaves zero. Problems 4–5 are the k = 2 case: IR 0.6 assumed, 0.3 true, VA falls from 0.19% possible to 0.

5. Optimal holdings (5A.26)–(5A.28)

```
h_P = \beta_P\, h_B + \frac{\mathrm{IR}}{2\lambda_R\,\omega_A}\, h_A =
\beta_P\, h_B + \frac{V^{-1}\alpha}{2\lambda_R}
```

- *Intuition:* the optimal active position is the alpha portfolio scaled to risk ω*. Beta is free because alphas are benchmark-neutral.

6. Alpha equals IR times marginal contribution to residual risk (5A.15)

```
\alpha = \mathrm{IR}\cdot\frac{V h_A}{\omega_A} = \mathrm{IR}\cdot
\mathrm{MCRR}_Q
```

- *Use (p. 136):* in an optimal portfolio with IR = 0.5, every alpha equals half its marginal contribution to residual risk. Regressing alphas on MCRR checks hand-built portfolios; the toolkit verifies this identity exactly.

7. Sharpe ratio, benchmark Sharpe ratio and IR (5A.16)

```
\mathrm{SR}^2 = \mathrm{SR}_B^2 + \mathrm{IR}^2
```

- *Example:* a benchmark with 6% expected excess return and 20% risk (SR_B = 0.30) plus alphas worth IR = 0.5 gives a maximum Sharpe ratio of 0.58.

8. The cost of no active beta and no active cash (5A.37)–(5A.42)

```
\mathrm{IR}_Y = \mathrm{IR}\sqrt{1 - \left(\frac{\mathrm{IR}_C}
{\mathrm{IR}}\right)^2} = \mathrm{IR}\sqrt{1 - \operatorname{Corr}^2\
{\theta_Q, \theta_C\}}
```

- *Intuition:* the constraints remove the part of your alpha that lines up with the residual of the minimum-variance portfolio. If alphas are tilted toward low-risk stocks (IR_C/IR = 0.3), the constrained IR is 95% of the unconstrained one.
- *Use:* a quick price tag on a full-investment rule. The toolkit checks this against a brute-force KKT solve.

9. Time scaling and alpha scaling (5A.50), Prop. 6

```
\frac{E\{\theta_{[0,T]}\}}{\operatorname{Std}\{\theta_{[0,T]}\}} =
\sqrt{\frac{T}{\Delta t}}\cdot\frac{\alpha}{\omega}, \qquad \mathrm{IR}
(\pi\alpha) = \pi\cdot\mathrm{IR}(\alpha)
```

- *Example (p. 144):* raw alphas implying $IR_0 = 2.46$ should be multiplied by 0.75/2.46 = 0.30 before optimizing with λ = 0.10.

D. Trading implications

1. **Size risk from IR and λ [book].** ω* = IR/2λ; with IR = 0.5 and a 4% risk target, λ = 0.0625.

2. **Scale alphas to a believable IR [book].** Compute $\sqrt{(\alpha^T V^{-1}\alpha)}$ first; if it exceeds about 1, shrink.
3. **Err toward modesty [derived].** Overstating IR by 2× destroys all value added; understating by 2× keeps 75%.
4. **Audit portfolios with α vs MCRR [book].** A low R^2 means positions do not express the alphas.
5. **Price every constraint in IR terms [book].** No-active-cash and no-active-beta via (5A.40); long-only via simulation (experiment E3).
6. **Rank strategies and managers by IR [book].** Allocate risk with ω* per strategy.

E. Practical implementation

```
import numpy as np, gk_part1 as gk

def build_active_portfolio(alpha_raw, V, h_B, target_IR=0.75,
target_te=0.04):
    b = gk.betas(V, h_B)
    alpha = alpha_raw - b * (alpha_raw @ h_B)              # benchmark-
neutral (p. 128)
    alpha = gk.scale_alphas_to_ir(alpha, V, target_IR)     # Prop. 6
    lam = gk.implied_residual_risk_aversion(target_IR, target_te)   # (5.11)
    h_P = h_B + gk.optimal_residual_position(alpha, V, lam) # (5A.26), beta =
1
    check = gk.alpha_mcrr_check(alpha, V, h_B, h_P)        # (5A.15): slope ~
IR, R^2 ~ 1
    return h_P, lam, check
```

F. Critical evaluation

- **The IR is assumed, then used.** The can-opener is honest but circular: every optimal policy depends on an IR you cannot observe and estimate with large error (±0.3 after 10 years). The overconfidence penalty above shows why this matters.
- **Constant IR across risk levels** fails with long-only constraints, capacity limits and costs (p. 116 admits it; E3 shows long-only IR falling from 0.98 to 0.75 as the risk target rises from 2% to 5%).
- **Empirical tables are short-window.** Mutual funds 1991–93, institutional portfolios 1995–96, with style-analysis benchmarks (Table 5.5). They mix skill, luck, and benchmark misfit.

- **Quadratic preferences** ignore skew and drawdown. For hedge-fund capital, the relevant IR is the Sharpe ratio (cash benchmark), and leverage limits cap ω^*.
- **The five-bin recipe** gives the same alpha to a 15% and a 60% volatility stock; Ch. 10 fixes this with alpha = volatility × IC × score.

G. Key takeaways

1. IR is the single number that describes opportunity: $\alpha = IR \times \omega$ along the frontier.
2. $\omega^* = IR/2\lambda$ and $VA^* = IR^2/4\lambda$; value added grows with the square of IR.
3. $IR = \sqrt{(\alpha^T V^{-1} \alpha)}$: squared IRs add across independent bets.
4. At the optimum, $\alpha = IR \times MCRR$; use it to audit portfolios.
5. Top-quartile before-fee IR is about 0.5; treat anything above 1 with suspicion.

Test your understanding

Conceptual questions

1. Why is the residual frontier a straight line, and what makes it bend in practice?
2. Why does every investor, regardless of risk aversion, prefer the highest-IR manager?
3. What does it mean that portfolio A has zero beta, and why does that require benchmark-neutral alphas?
4. Why does the monthly IR equal the annual IR divided by $\sqrt{12}$, and what assumption could break it?
5. Why is overestimating IR more costly than underestimating it by the same factor?

Math exercises

1. What IR delivers 2.5% value added at $\lambda_R = 0.10$, and how much residual risk does it require (Problem 2)?
2. With Q = MMI, $f_Q = 6\%$, B = cap-weighted MMI, and the betas and risks on p. 133, compute f, α, ω, SR and IR for B, Q and C (Problem 3).
3. $\lambda_R = 0.12$ and IR = 0.6: find ω^* and VA^*. If the true IR is 0.3, how much value added is lost (Problems 4–5)?

Solutions. (1) $IR^2 = 4 \times 0.10 \times 2.5 = 1.0$, so IR = 1.0 and $\omega^* = 5\%$. (2) $f_B = 6 \times 0.965 = 5.79\%$, $f_C = 6 \times 0.831 = 4.99\%$; $\alpha_Q = 6 - 1.004 \times 5.79 = 0.19\%$, $\alpha_C = -0.02\%$; $\omega_Q = \sqrt{(15.82^2 - 1.004^2 \times 15.5^2)} = 2.85\%$, $\omega_C = 5.31\%$; $SR_B = 0.374$, $SR_Q = 0.379$, $SR_C = 0.346$; $IR_Q =$

0.066, IR_C ≈ 0, IR_B = 0. Check: $\sqrt{}(0.379^2 - 0.374^2) = 0.066$, as (5A.16) requires. (3) $\omega^* = 2.5\%$, $VA^* = 0.75\%$. With true IR 0.3 at $\omega = 2.5\%$, VA = 0.75 − 0.75 = 0; the optimum would have been 0.19%, so 0.19% is lost.

Coding exercises

1. Implement the five-bin direct-alpha recipe (p. 128) and compare its IR, via $\sqrt{}(\alpha^T V^{-1}\alpha)$, with a volatility-scaled version ($\alpha = \omega \times IC \times score$) on a factor-model V.
2. Plot value added against risk for true IR = 0.5 and assumed IR from 0.25 to 1.25, showing the $(2k - k^2)$ curve.

Research experiment

For each of your live or paper strategies, compute the ex-ante IR implied by its alphas ($\sqrt{}(\alpha^T V^{-1}\alpha)$) at each rebalance and the realized IR over the following 12 months. The ratio realized/ex-ante is your empirical overconfidence factor k; use it to rescale alphas going forward, and test whether it is stable across years.

Connections: the value-added objective comes from Ch. 4, the alpha portfolio is Ch. 2's characteristic-portfolio machinery, and MCRR comes from Ch. 3. Chapter 6 opens the can: it explains where IR comes from.

10. Chapter 6 — The Fundamental Law of Active Management (pp. 147–172)

A. Central thesis

A strategy's information ratio is approximately its skill per bet times the square root of the number of independent bets per year, $IR \approx IC\sqrt{BR}$. A small edge applied often and across many assets beats a large edge applied rarely (p. 162: "you must play often and play well"). Squared IRs add across independent sources, which gives a way to budget research and to combine strategies.

B. Key concepts

- **Breadth BR (p. 148):** the number of independent forecasts of exceptional return per year. Not the number of names, not the number of rebalances.
- **Information coefficient IC (p. 148):** the correlation between each forecast and the realized outcome, assumed equal across forecasts.

- **Trade-offs (p. 149):** a 50% increase in breadth equals a 22% increase in skill. A quarterly market timer needs IC = 0.25 for IR = 0.5; a stock picker covering 100 stocks quarterly needs IC = 0.025.
- **Three examples (pp. 150–154):** roulette from the casino's side; 200 stocks whose residual returns are sums of 300 equal "nuggets", of which you know one; and directional market calls.
- **Additivity (pp. 154–157):** squared IRs add across stock groups, across timing and selection, across currency, country and stock decisions, and across managers (if the sponsor allocates optimally).
- **Assumptions (pp. 157–160):** independent sources, equal skill, and above all a manager who knows the true value of his or her information and uses it optimally.
- **Not the law of large numbers (p. 160):** the law holds at BR = 10 as well as at BR = 1,000; breadth diversifies residual risk, it does not shrink sampling error.
- **Tests (pp. 160–161):** in simulations with alphas mixed to a target IC, realized IRs of optimal portfolios matched the law; short-sale limits pushed them down.
- **Investment style (p. 161):** the law favours eclectic managers. In practice managers specialize and "the sponsor makes the stew".

C. Mathematical foundations

1. The law, aggressiveness and value added (6.1)–(6.3)

$$\mathrm{IR} \approx \mathrm{IC}\sqrt{\mathrm{BR}}, \qquad \omega^* = \frac{\mathrm{IC}\sqrt{\mathrm{BR}}}{2\lambda_R}, \qquad \mathrm{VA}^* = \frac{\mathrm{IC}^2\,\mathrm{BR}}{4\lambda_R}$$

- *Intuition:* each bet contributes IC² of squared IR, like independent random variables contributing variance. Value added is linear in breadth and quadratic in skill.
- *Examples (p. 149):* 0.25√4 = 0.5 (quarterly timer); 0.025√400 = 0.5 (100 stocks quarterly, or 2 stocks revised 200 times a year). √1.5 = 1.22, so +50% breadth = +22% skill.
- *Assumptions:* independent bets, equal IC, optimal use of information, IC small (the approximation drops a variance-reduction term).

2. Where the approximation comes from (6A.4)–(6A.21)

$$E\{\theta\mid z\} = A\,P\,J\,z,\quad \operatorname{Var}\{\theta\mid z\} = A(I - PP^{\top})A^{\top},\quad \mathrm{IR}^2 = \operatorname{Tr}\{P^{\top}(I-PP^{\top})^{-1}P\} \approx \sum_{n,b}\rho_{n,b}^2 = \mathrm{BR}\cdot\mathrm{IC}^2$$

- *Variables:* residual returns θ = Ax with x orthonormal; signals z = Ey with y orthonormal ($J = E^{-1}$); P is the correlation matrix between x and y.
- *Steps:* (i) conditional mean and variance of θ given the signals (best linear unbiased estimate); (ii) optimal residual holdings $h^* = (1/2\lambda)\, G^{-1}\,\alpha(z)$ with G the conditional covariance; (iii) conditional squared IR = $y^{T}P^{T}DPy$ with $D = (I - PP^{T})^{-1}$; (iv) average over y; (v) expand $D = I + PP^{T} + \ldots$ and keep the first term because correlations are small.
- *Exact version (derived here):* when each signal is informative about its own orthogonal component with correlation IC, PP^{T} has BR eigenvalues IC^2, so $IR^2 = BR \times IC^2 / (1 - IC^2)$. This reproduces the book's "actual" numbers to four decimals: the casino's 0.027037 and the nugget example's 1.1566, where the law gives 0.027027 and 1.154.

3. Casino (pp. 150–151)

- One $1 bet with a 19/37 chance of winning: expected return 2.7027%, standard deviation 99.9634%, IR = 0.02704 versus $IC\sqrt{BR} = (1/37)\sqrt{1} = 0.02703$.
- One million bets a year: IR = 27.04 versus 27.03. Two green slots (IC = 2/38): 52.70 versus 52.63.

4. The nugget model (pp. 151–153)

$$\theta_n = \sum_{j=1}^{300}\theta_{n,j},\quad \theta_{n,j} = \pm 1\%,\qquad \mathrm{IC} = \frac{1}{\sqrt{300}} = 0.0577,\qquad \mathrm{IR} = 0.0577\sqrt{400} = 1.154$$

- *Portfolio:* equal-weight benchmark of 200 stocks; buy list at 1% each, sell list at 0%. Expected active return 1% per half-year, active risk $\sqrt{(299/200)} = 1.2227\%$, IR = 0.8179 per half-year or 1.1566 a year.
- *With costs (fn. 4, p. 153):* about 100% turnover a year at 0.80% round trip cuts IR to 0.70. Costs consume 40% of a top-decile IR.

5. Directional forecasts (6.5)–(6.6)

$$\mathrm{IC} = \frac{1}{N}\sum_t x(t)\,y(t) = \frac{N_1 - (N - N_1)}{N} = 2\frac{N_1}{N} - 1$$

- *Derivation:* with x, y = ±1 (mean 0, variance 1), the correlation is the average product, +1 for a hit and −1 for a miss.
- *Examples (p. 154):* IC = 0.0577 is a 52.9% hit rate; IC = 0.02 is 51%, and over 200 stocks quarterly gives IR ≈ 0.57.

6. Additivity (6.7)–(6.8) and combining managers (p. 157 fn. 7)

$$\mathrm{IR}^2 = \mathrm{BR}_1\,\mathrm{IC}_1^2 + \mathrm{BR}_2\,\mathrm{IC}_2^2, \qquad y_n^* = \frac{\mathrm{IR}_n}{2\lambda_{SA}\,\omega_n} \;\Rightarrow\; \mathrm{IR} = \sqrt{\textstyle\sum_n \mathrm{IR}_n^2}$$

- *Examples:* 200 stocks semiannual at IC 0.04 gives 0.8; adding 100 stocks at IC 0.03 adds 0.18 to IR², for IR 0.906 and 28% more value added (p. 154). Stock selection (0.6) plus quarterly timing at IC 0.1 (0.2) gives 0.63 (p. 155). Three managers with IR 0.75, 0.50, 0.30 combine to 0.95 (p. 157).
- *International (6.8):* 1,000 stocks at IC 0.02 contribute 0.40 to IR²; currency and country decisions (24 bets a year) need IC 0.129 to match, and at a more realistic 0.075 add only 0.135, for a total IR of 0.73 (p. 156).

7. Correlated sources (6.9)

$$\mathrm{IC}(\text{com}) = \mathrm{IC}\sqrt{\frac{2}{1+\gamma}}$$

- *Derivation:* the best linear combination of two standardized signals with covariances c = (IC, IC) to the return and correlation matrix S = [[1, γ], [γ, 1]] has squared correlation $c^T S^{-1} c = 2IC^2/(1 + \gamma)$.
- *Example (p. 159):* two sources each knowing two nuggets, sharing one (γ = 0.5): each has IC 0.0816 and together 0.0942, not $0.0816\sqrt{2} = 0.115$.

D. Trading implications

1. **Price research projects with the law [book, p. 149].** Compare the IR² a project adds (new BR × IC²) with its cost.
2. **Favour cross-sectional over timing strategies [book].** Timing breadth comes only from frequency (Ch. 19).
3. **Measure effective breadth [book + inference].** Regress signals on industry and style exposures (p. 158), and treat rebalancing on unchanged information as one bet.
4. **Combine weak, uncorrelated signals [book].** Each additional independent signal adds IC² × BR to IR².

5. **Expect less than the law [inference + E3].** Costs (fn. 4), constraints (p. 161) and IC instability all cut realized IR.

E. Practical implementation

```
import numpy as np, gk_part1 as gk

def research_budget(projects, ir_now):
    """projects: list of dicts with 'name', 'IC', 'BR', 'corr_with_existing'
(0..1)."""
    rows = []
    for p in projects:
        add = p['BR'] * p['IC']**2 * (1 - p['corr_with_existing']**2)   #
crude haircut for overlap
        rows.append((p['name'], round(np.sqrt(ir_now**2 + add) - ir_now, 3)))
    return sorted(rows, key=lambda r: -r[1])

def effective_breadth(signal_corr, ic):
    """Signals with common IC and correlation matrix S: IR^2 = ic^2 * 1'S^-1
1, so BR_eff = 1'S^-1 1."""
    one = np.ones(len(signal_corr))
    return float(one @ np.linalg.solve(signal_corr, one))
```

Experiments E1 and E2 in the toolkit (Section 12) simulate these formulas: realized IRs matched IC√BR within one standard error, and correlated signals matched (6.9).

F. Critical evaluation

- **Hypercompetence is the weak link** (p. 162). Real ICs are estimated with error, drift over time, and differ across sectors and regimes.
- **The law ignores IC volatility over time [outside: Qian and Hua, 2004].** For large universes, realized IR is closer to mean(IC) / std(IC) × √(periods per year). Cross-sectional breadth is capped by how much IC moves from month to month.
- **Implementation efficiency is missing [outside: Clarke, de Silva and Thorley, 2002].** The "transfer coefficient" TC extends the law to IR = TC × IC × √BR; long-only constraints typically push TC well below 1 (E3's IR loss is this effect).
- **Breadth is undefined operationally.** Factor structure, sector concentration and persistent signals all cut effective breadth well below N × rebalances.
- **The tests are simulations of the law's own model** (p. 161), not market evidence. They show the algebra is right, not that markets offer the assumed independent bets.

- **Additivity across managers** needs optimal sponsor allocation and uncorrelated active returns; real manager line-ups share style bets.

G. Key takeaways

1. IR ≈ IC√BR; value added ∝ IC² × BR.
2. Breadth means independent bets per year, not stocks × rebalances.
3. Tiny ICs (0.02–0.05, hit rates of 51–52.5%) are enough with real breadth.
4. Squared IRs add across independent sources; correlated sources add much less.
5. Treat the law as an upper bound to be reduced by costs, constraints and IC instability.

Test your understanding

Conceptual questions

1. Why does rebalancing monthly on an annual view not add breadth?
2. How can an analyst's 50 stock picks be a single bet?
3. Why is the fundamental law not the law of large numbers?
4. What happens to the law when skill varies across stocks (Fig. 6.3)?
5. Why do sponsors, not managers, end up building eclectic portfolios?

Math exercises

1. Manager A follows 250 stocks quarterly with IC 0.02. Manager B times four equity styles quarterly. What IC must B have to match A, and what IR does the sponsor get if B's IC is 0.08 (Problem 1)?
2. A manager covers 500 stocks monthly with IC 0.05 and IR 1.0. How many bets and how many independent bets does he make (Problem 2)?
3. Compare three research programs on the nugget model: (a) 200 stocks quarterly knowing two nuggets, (b) the same with two different nuggets, (c) 100 stocks quarterly knowing three nuggets (Problem 3).

Solutions. (1) IR_A = 0.02√1000 = 0.632. Treating B's 16 quarterly style calls as independent, IC_B = 0.632/4 = 0.158; with IC_B = 0.08, IR_B = 0.32 and the combination is √(0.40 + 0.10) = 0.71. Because style bets are relative, only three of four may be independent; then IC_B would need to be 0.183. (2) 6,000 forecasts but only $(1.0/0.05)^2$ = 400 independent ones: his alphas are highly persistent or correlated. (3) IC = √(k/300) for k known nuggets. (a) and (b) tie at IR^2 = 800 × 2/300 = 5.33; (c) gives 400 × 3/300 = 4.0. Breadth wins.

Coding exercises

1. Extend E1 so IC varies each quarter, IC_t ~ N(0.03, 0.03^2). Show realized IR approaches mean(IC)/std(IC) × 2 as N grows, not $0.03\sqrt{(4N)}$.
2. Given 10 signals with known ICs and a correlation matrix, compute optimal combination weights and the combined IC $\sqrt{(c^T S^{-1} c)}$, and compare with the naive equal-weight blend.

Research experiment

For 3–5 standard US equity signals (value, 12-1 momentum, short-term reversal, earnings revisions, quality), measure the monthly cross-sectional IC over 20 years. For each signal and for their optimal blend, compare: (a) $IC\sqrt{(12N)}$, the naive law; (b) $IC/\text{std}(IC) \times \sqrt{12}$; (c) the realized IR of an optimal unconstrained portfolio; (d) the realized IR long-only. The gaps between (a), (b), (c) and (d) quantify lost breadth, IC instability and constraint costs.

Connections: Ch. 5 assumed the IR; this chapter decomposes it. The appendix uses Ch. 2's portfolio machinery and Ch. 4's value-added objective. The law drives the skepticism about benchmark timing (Ch. 19), the high bar for asset allocation (Ch. 18), the forecasting rule alpha = volatility × IC × score (Ch. 10), and information analysis (Ch. 12).

11. Strategy catalog (Part One entries)

Part One contains no data-driven trading rules; it contains five strategy templates that the authors use to illustrate the framework, plus one outside hypothesis that the CAPM chapter invites. In each table, **Book** is what the authors specify (with page), and **Proposed** is my addition, to be tested, not attributed to them. Later installments add the signal-level strategies (DDM, comparative valuation, momentum, residual reversal, estimate revisions, event studies).

S1. Ranked stock selection with direct alphas and an optimal residual position

Field	Book	Proposed
Universe	Benchmark constituents; examples use MMI and S&P 500 (pp. 127–128)	Russell 1000, point-in-time membership
Source of alpha	Any ranking with modest skill; skill measured by IC (Ch. 6)	The ranking needs its own economic story (Part Two)

Field	Book	Proposed
Data	Rankings, betas, a covariance model	Survivorship-free prices with delisting returns, PIT fundamentals, factor risk model
Features	Five bins at +2/+1/0/−1/−2% alpha, then subtract benchmark-average alpha × beta (p. 128)	Scale by residual volatility and IC (preview of Ch. 10), then to a target IR (Prop. 6)
Entry	Hold $h_B + V^{-1}\alpha / 2\lambda$ (5A.26)	Re-solve when rankings update
Exit	Implicit: positions follow the alphas	No-trade band sized by the cost model
Sizing	$\omega^* = IR / 2\lambda$; λ from the tracking-error target (5.10)–(5.11)	IR 0.5, 4% target gives $\lambda = 0.0625$
Risk management	$\beta = 1$; check α against MCRR (5A.15)	Neutralize industry and size unless intended; stress at high market volatility
Frequency	Not specified; examples quarterly or semiannual	Monthly rebalance
Costs	Set aside in Ch. 5 (p. 109)	Spread + square-root impact, turnover penalty (Ch. 16)
Backtest	Not covered in Part One (see Ch. 12)	Walk-forward, PIT data, purged IC estimation
Metrics	IR, value added, active risk	Plus IC series, turnover, cost drag, transfer coefficient
Regimes	—	Depends on the ranking source
Overfitting risk	—	Bin thresholds and IC estimated in-sample
Invalidation	—	Out-of-sample IC ⩽ 0, or α-vs-MCRR R^2 persistently low

S2. Buy list / sell list, long-only, equal weights (nugget example)

Field	Book	Proposed
Universe	200 stocks, equal-weighted benchmark at 0.5% each (p. 152)	Any equal-weight universe; cap-weighted needs Ch. 15's adjustments
Source of alpha	Knowledge of 1 of 300 equal return components (IC = 0.0577)	A real signal with measured IC
Data	Forecast sign per stock	Same plus residual-return history to estimate IC
Features	Forecast residual return +1% (buy) or −1% (sell)	Use refined alphas instead of signs
Entry	Buy list held at 1% each (p. 152)	Same
Exit	Stock moves to sell list → sold to 0%	Add hysteresis to cut turnover
Sizing	Equal 1% (twice benchmark weight)	Volatility-scaled weights
Risk management	Active risk 1.22% per half-year (p. 153)	Factor-neutral buy and sell lists
Frequency	Semiannual, about 50% turnover per half-year (fn. 4)	Match to signal half-life (Ch. 13)
Costs	0.80% round trip cuts IR from 1.16 to 0.70 (fn. 4)	Name-specific cost estimates
Backtest	Analytic (synthetic model)	Event-time and calendar-time backtests
Metrics	IR, active return	Hit rate, IC, turnover
Regimes / overfitting / invalidation	Not discussed	List membership is a coarse signal; test sensitivity to the buy/sell cutoff

S3. Benchmark timing through active beta

Field	Book	Proposed
Universe	The benchmark itself (Ch. 4; full treatment in Ch. 19)	Index futures on the benchmark
Source of alpha	Forecast of exceptional benchmark return Δf_B	Valuation or macro signals (to be tested)
Entry and sizing	$\beta_{PA} = \Delta f_B / (2\lambda_{BT} \sigma_B^2)$ (4.16); example ±0.05 beta moves each quarter (p. 158)	Cap absolute active beta at 0.1
Breadth and skill	Quarterly calls: IC 0.25 needed for IR 0.5 alone (p. 149); IC 0.1 adds IR 0.2 (p. 155)	Raise frequency only with genuinely new information
Exit	Next forecast	Same
Risk / costs	Separate aversion λ_{BT}; futures preferred to stocks (Ch. 19 intro)	Roll and margin costs
Metrics	Timing value added, first term of (4.15)	Timing regression (Ch. 17–19)
Invalidation	—	Out-of-sample IC ⩽ 0 over 40+ independent calls

S4. Layered international strategy: currency, country, stock

Field	Book	Proposed
Universe	London-based investor; US, Japan, Germany, UK (pp. 155–156)	Developed-market equity indices and stocks
Structure	3 currency bets and country allocation, each quarterly (12 + 12 bets a year); stocks: 400 US, 300 Japan, 200 UK, 100 Germany, revised yearly	Same, with separate risk budgets

Field	Book	Proposed
Skill assumptions	Stocks IC 0.02 ($IR^2 = 0.40$); currency and country IC 0.075 ($IR^2 = 0.135$); total IR 0.73 (6.8)	Measure each IC separately
Sizing	Optimal combination (squared IRs add)	ω^* per layer from IR_layer / 2λ
Lesson	Timing layers need IC 0.129 to match 1,000 stocks at 0.02	Fund research where BR × IC^2 per dollar is highest
Other fields	Not specified	As S1

S5. Combining managers or strategies by IR

Field	Book	Proposed
Allocation rule	$y_n = IR_n / (2\lambda_{SA} \omega_n)$, giving $IR = \sqrt{(\Sigma IR_n^2)}$ (p. 157 fn. 7)	Shrink each IR_n toward the population mean before allocating
Example	IR 0.75, 0.50, 0.30 combine to 0.95 (p. 157)	—
Assumption	Independent active returns, optimal allocation	Estimate the correlation of active returns; use $\sqrt{(IR^T C^{-1} IR)}$ instead
Invalidation	—	Realized combined IR far below the formula, which signals shared style bets

S6. Low-beta tilt (outside the book; hypothesis)

Field	Proposed
Motivation	If the realized security market line is flatter than the CAPM line (Black 1972, cited p. 25; Frazzini and Pedersen 2014), consensus $\beta\mu_B$ overstates high-beta returns
Signal	Negative shrunk beta, sector-neutral

Field	Proposed
Construction	Beta-neutral long/short via (5A.26) with β = 0 active, or long-only tilt
Main risks	Leverage needed to neutralize beta; rate sensitivity; crowding
Invalidation	No significant alpha after costs and after controlling for value, quality and size

12. Python implementation guide (Part One)

The toolkit implements every Part One formula as a function tagged with its equation number, reproduces the book's worked numbers in 34 passing tests, and runs four experiments on simulated data. The files are attached to the chat (`gk_part1.py` , `test_gk_part1.py` , `experiments_part1.py` , `results_part1.json` , `README.md`).

Chapter	Functions	Equations
2	`ols_alpha_beta` , `characteristic_portfolio` , `portfolio_C` , `portfolio_q_Q` , `implied_excess_returns` , `efficient_frontier_variance`	(2.2), (2A.1)–(2A.3), (2A.14), (2A.23)–(2A.24), (2A.36), (2A.47)
3	`FactorRiskModel` (cov, Woodbury inverse, factor portfolios, decomposition), `gls_factor_returns` , `marginal_contributions` , `active_risk_attribution` , `equal_corr_portfolio_vol`	(3.4), (3.6)–(3.8), (3A.2)–(3A.4), (3A.20)–(3A.23), (3A.31)–(3A.38)
4	`decompose_expected_return` , `total_risk_aversion_consistent_with_benchmark` , `optimal_active_beta` , `value_added`	(4.7)–(4.9), (4.12), (4.15)–(4.16)
5	`information_ratio_from_alphas` , `portfolio_A` , `optimal_residual_position` , `portfolio_Y` , `alpha_mcrr_check` , `scale_alphas_to_ir` , `optimal_residual_risk` , `optimal_value_added`	(5.10)–(5.12), (5A.1)–(5A.7), (5A.15), (5A.26), (5A.37)–(5A.42), Prop. 6

Chapter	Functions	Equations
6	`fundamental_law_ir` (with exact variant), `combine_information_ratios`, `ic_from_hit_rate`, `combined_ic`, `refined_alpha_forecast`	(6.1), (6.6)–(6.9), (6A.14), p. 161 fn. 8

```
pip install numpy scipy cvxpy pytest
python -m pytest -q test_gk_part1.py      # 34 passed
python experiments_part1.py               # about 10 seconds
```

The tests found one last-digit rounding in the text: the casino IR printed as 0.027038 is 0.027037 when recomputed (p. 151). Everything else in Part One that can be recomputed matched.

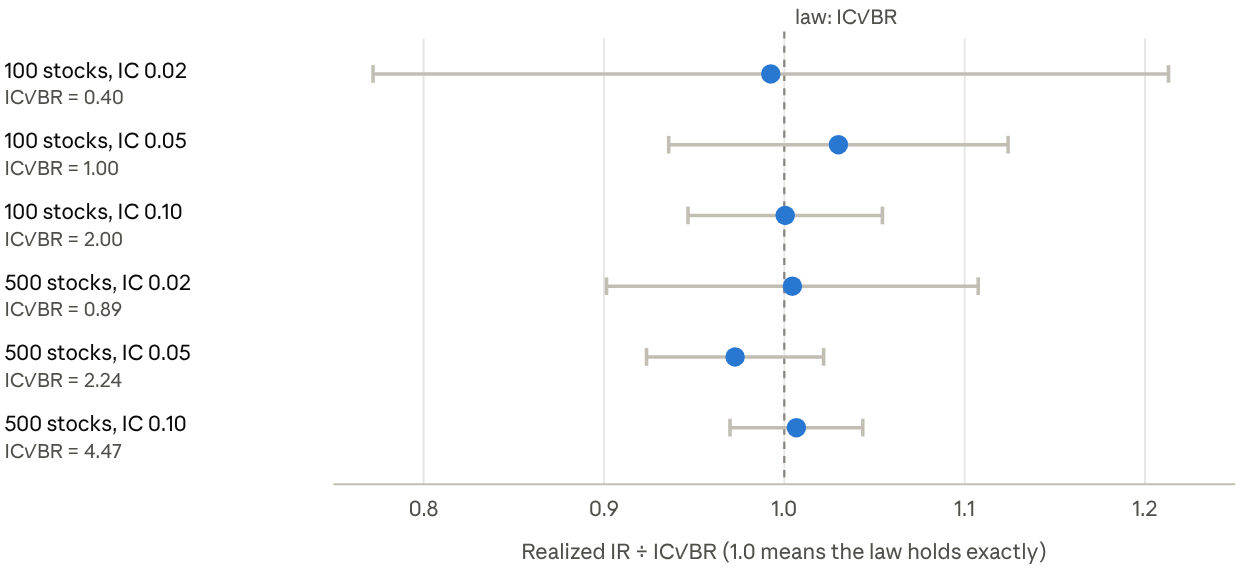
Experiment results (simulated data only)

All four experiments draw returns from known models, so they test the mathematics and common implementation traps. They are not evidence about real markets; Section 13 lists the real-data versions.

E1 — Does IR ≈ IC√BR hold for optimal portfolios? Independent residual returns (20–40% volatility), forecasts built with the book's fn. 8 recipe, optimal unconstrained positions, 2,000 quarters per setting. All six realized IRs landed within 1.1 standard errors of the law; the chart below plots them.

Realized IR matched IC√BR in all six simulated settings

Realized ÷ predicted IR, 2,000 simulated quarters each; whiskers are 95% intervals; largest gap 1.1 standard errors



experiments_part1.py, E1 · simulated data, 6 settings × 2,000 quarters

Every 95% interval contains 1.0, so at ICs up to 0.10 the law's approximation error is invisible next to sampling noise.

E2 — Correlated signals. Two signals with IC 0.04 each, 500 stocks quarterly, optimally combined; mean of six seeds.

Signal correlation γ	Realized IR (± sd across seeds)	(6.9) prediction	If counted as independent
0.0	2.53 ± 0.04	2.53	2.53
0.5	2.07 ± 0.04	2.07	2.53
0.9	1.83 ± 0.03	1.84	2.53

At γ = 0.9 the second signal adds almost nothing: one signal alone gives 1.79.

E3 — Long-only constraint. 300 stocks, skewed cap-weighted benchmark, IC 0.03 quarterly (law IR 1.04), 800 quarters, same risk aversion for both books.

Risk target	IR long/short	IR long-only	Ex-ante TE long/short	Ex-ante TE long-only	Long-only active weight vs log cap
2%	0.96	0.86	2.0%	1.4%	correlation −0.25

Risk target	IR long/short	IR long-only	Ex-ante TE long/short	Ex-ante TE long-only	Long-only active weight vs log cap
5%	0.98	0.75	5.0%	2.6%	correlation −0.30

The long-only book cannot reach its risk target, loses more IR as it tries to be more aggressive, and tilts toward small caps: the Ch. 5 "IR independent of aggressiveness" assumption breaks exactly as p. 116 warns, and Ch. 15's size bias appears.

E4 — Risk model error. 100 stocks generated by a 6-factor model; risk estimated from 60 or 120 months; predicted versus realized risk under the true covariance (median of 200 portfolios).

History	Model	Optimized portfolios: realized / predicted	Random portfolios: realized / predicted
60 months	Estimated factor model	1.08	1.02
60 months	Sample covariance (rank 59 of 100)	A zero-net position with predicted risk ≈ 0 has true risk 4.2%	—
120 months	Estimated factor model	1.05	1.02
120 months	Sample covariance	5.57	1.03

The sample covariance looks fine on random portfolios and fails badly on the portfolios an optimizer actually picks (p. 54).

13. Research questions and experiment backlog

The ten experiments below turn Part One's claims into tests on real data, ranked by how much they would change a research decision relative to their cost. AlphaStockPro's proprietary stock scores are a ready-made signal for experiments 1, 3 and 4: they test whether the scores carry information and how much of it survives implementation.

#	Experiment	Claim tested (source)	Data needed	Priority	Status
1	Breadth efficiency: realized IR versus IC√BR for 3–5 signals and their blend	Fundamental law (6.1); Qian–Hua IC-volatility bound (outside)	Survivorship-free daily prices, PIT fundamentals, 15+ years	P1	Not started
2	Risk-model bias statistics: sample, shrinkage, statistical and fundamental models on optimized portfolios	Structural models forecast risk without bias (pp. 67–70); E4	Daily returns, factor exposures	P1	Not started
3	Overconfidence factor k = realized IR / ex-ante $\sqrt{(\alpha^T V^{-1}\alpha)}$ per strategy and year	Value added falls as $(2k - k^2)$ (derived from (5.9))	Live or paper alphas and holdings	P1	Not started
4	Effective breadth of a multi-signal book: $1^T S^{-1} 1$ from the signal correlation matrix versus raw count	(6.9) and p. 158 independence warning	Signal panels	P1	Not started
5	Long-only versus long/short IR and size tilt on the same alphas	IR independent of aggressiveness (p. 116); E3; Ch. 15 preview	As 1, plus cap weights	P2	Not started
6	Alpha-vs-MCRR audit of existing portfolios (slope = implied IR, R^2 = consistency)	(5A.15), p. 136	Holdings, alphas, risk model	P2	Not started
7	Beta forecasting: OLS versus Bayesian-shrunk versus fundamental betas, next-60-month error	Betas regress toward 1 (p. 14 fn. 4)	Monthly returns, fundamentals	P2	Not started
8	Diversification under stress: realized average correlation and tracking error versus market volatility	(3A.40)–(3A.41)	Daily returns 2000–2026 incl. 2008, 2020	P2	Not started
9	Low-beta tilt net of costs and factor controls	Outside hypothesis S6	As 1	P3	Not started

#	Experiment	Claim tested (source)	Data needed	Priority	Status
10	Sell-side target decomposition: IC of the residual alpha after stripping (4.7) components	Exceptional return is all that matters (Ch. 4)	Analyst price targets, PIT	P3	Not started

Open research questions

- How should effective breadth be estimated when bets share factor exposure and signals persist across months?
- Is IC forecastable from regime variables (dispersion, volatility, crowding), and does conditioning raise IR or only add noise?
- Does equal skill across sectors and capitalization bands hold for any real signal, or should the law be applied bucket by bucket (Fig. 6.3)?
- What does the IR-versus-risk curve look like for a real long-only mandate once costs and position limits are included?
- Is variance enough as a risk measure for crowded factor strategies, or does an explicit crowding or tail term earn its complexity?
- Is a reverse-optimized benchmark the right neutral prior, or does a factor-model prior (Ch. 7) forecast better?